

Credit Risk Analysis: Retail Mortgage Loans

State-Dependent Logistic Hazard Model for Short-Term Default Prediction

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Overview

We create a 1-month-ahead probability of default (PD) model using logistic regression with panel data comprised of loan origination characteristics, monthly loan performance variables and macroeconomic variables. That is, using month-level data, we use information available at time t (including information from origination) to predict default at time $t + 1$. We use loan-clustered robust standard errors to account for unobserved heterogeneity. One of the covariates we used is the rolling count of delinquencies (missed payments) up to the current time, which is effectively what makes the regression a state-dependent model¹. Due to the inclusion of these covariates, the model is best interpreted as an early-warning PD model useful for credit portfolio monitoring (e.g. detect loans at high risk of default) and account management (e.g. offer deals to stable borrowers)².

Why Logistic Regression?

Tree-based or ML models could potentially improve raw predictive performance, but for this application the marginal gains are often outweighed by reduced interpretability, stability, and governance complexity. Logistic regression provides a strong baseline that aligns with the hazard structure and allows clear economic interpretation.

Censoring

We face natural right-censoring at the data cutoff date (the event status in month $T + 1$ is unobserved for the final observed month T). In addition, to enforce an absorbing default state under our simplified default definition (3+ missed payments), we truncate each loan's panel after its first default month. Due to this, we implicitly condition on the loan surviving up to the current time, so we can also view the regression as a hazard rate model.

Data Source

We form a sample by combining data from 2013, 2016 and 2020 vintages sourced from the Freddie Mac Single Family Loan-Level Dataset (SFLLD) which can be found at SFLLD Site. These

¹In fact, the model could be extended by fitting the same regression to other definitions of default using the other states (1 missed payment, 2 missed payments), which would make that collection of models able to represent a 3-state time- and loan-dependent Markov transition matrix.

²This is not a model for the probability that a loan ever defaults using using origination data only, which would be a tool more useful for loan pricing than ongoing portfolio management.

origination years are chosen to provide a sample of loans originating in a variety of macroeconomic environments and to help identify the possible effects of loan age³ separately from calendar time.

Model Development Workflow

Predictors for next-month PD are selected based on economic justification and empirical assessment of their relationship with the log-odds of default. For correlated covariates, we assess multicollinearity and retain a single representative predictor to improve model stability and interpretability. After variable selection, we adopt a time-based 70/30 split, training the model on observations from February 2013 to September 2021 and evaluating performance on a held-out test set covering October 2021 to May 2025.

Model Definition

We define the event of default as 90+ days past due (3+ missed monthly payments). Some borrowers never make another monthly payment after “default”, although there are also some⁴ borrowers that subsequently cure (return to < 90 days past due) after “defaulting” and then possibly “default” again. We wish to define default as an absorbing state, meaning that a lender remains in default once they default. To enforce this, for each loan we drop the rows for months following the month of default. This simplified definition allows us to have more cases of “default” than if we only included actual defaults which never cured.

Also note that we treat prepayment as a right-censoring event. Specifically, once a loan prepays, it permanently exits the risk set for default, and we accordingly drop the observation corresponding to the prepayment month as well as all subsequent months for that loan. This is the approach of modeling cause-specific default hazard, treating prepayment as an independent⁵ censoring mechanism rather than explicitly modeling competing risks.

We will fit the following logistic regression model (Binomial GLM with canonical link):

$$(Y_{it}|X_{it}, Z_t) \sim \text{Bernoulli}(\pi_{it})$$

$$\text{logit}(\pi_{it}) = X_{it}^T \beta^{(L)} + Z_t^T \beta^{(M)}$$

where:

- $\text{logit}(p) = \ln(p/(1-p))$ is the logistic ("log odds") function,
- $Y_{it} = \mathbb{1}[\text{loan } i \text{ defaults in month } t+1]$,
- $\pi_{it} = \mathbb{P}(Y_{it} = 1|X_{it}, Z_t) = \text{PD}_{it}^{\text{1mo}}$ is the probability of default in month $t+1$ (given survival up to time t and the covariates for loan i up to time t),
- X_{it} is a vector of covariates for loan i at month t ,
- $\beta^{(L)}$ is the parameter vector for the loan-specific associations,
- Z_t is a vector of exogenous macroeconomic covariates at month t (appropriately lagged so that they are observed before t), and

³Loan age is the natural measure of "time" for a specific loan.

⁴Approximately 10% of loans in our data have this property.

⁵When we say independence here, we mean that we assume prepayment and default are conditionally independent given the covariates in the model.

- $\beta^{(M)}$ is the parameter vector for the macroeconomic associations.

Hazard Model View

This can also be viewed as a discrete hazard model:

Let $T_i \in \{1, 2, 3, \dots\}$ be the month of default for loan i . The covariate-dependent hazard (instantaneous risk) function for loan i at month $t + 1$ is

$$\begin{aligned} h_i(t + 1 | X_{it}, Z_t) &= \mathbb{P}(T_i = t + 1 | T_i \geq t + 1, X_{it}, Z_t) \\ &= \mathbb{P}(\text{default in month } t + 1 | \text{no default as of month } t, X_{it}, Z_t) \\ &= \mathbb{P}(Y_{it} = 1 | X_{it}, Z_t) \\ &= \pi_{it} \end{aligned}$$

This interpretation is valid because we removed rows for loans in the months after they defaulted⁶ and we dropped the final month of censored observations⁷, so each loan-month observation represents a period during which the loan is actually at risk of default.

Independence and Inference

We use the standard GLM likelihood (assuming full independence) as a working model for the mean response (one-month ahead default probability). Cross-loan independence is a plausible assumption after conditioning on macro covariates, although residual within-loan dependence may remain due to unobserved borrower and servicing effects. Under correct specification of the conditional mean $\mathbb{E}(Y_{it} | X_{it}, Z_t) = \pi_{it}$, the MLE remains a good estimator, so point estimation and prediction are okay. However, naive standard errors can be biased under un-modeled within-loan dependence, so for confidence intervals we use loan-clustered robust standard errors (CRSEs).

Data Background

Loan Data

We are concerned with modeling the event of default in the *next* month using information available in the *current* month.

The official document with the full data dictionary and user guide for SFLLD can be found at user guide.

There exist variables which may be useful that we did not use, such as “Estimated LTV” in the monthly performance data which relies on Freddie Mac’s proprietary Automated Valuation Model. These variables are omitted so that all variables in the final model are those which are readily available at a typical bank or credit union.

We drop loans with NA values for LTV at origination since it is an extremely small portion of our sample and none of those loans end up defaulting.

For loans with NA values for DTI at origination (`orig_dti`), there are actually a material amount which end up defaulting, so we can’t simply drop these loans. We create an indicator variable called

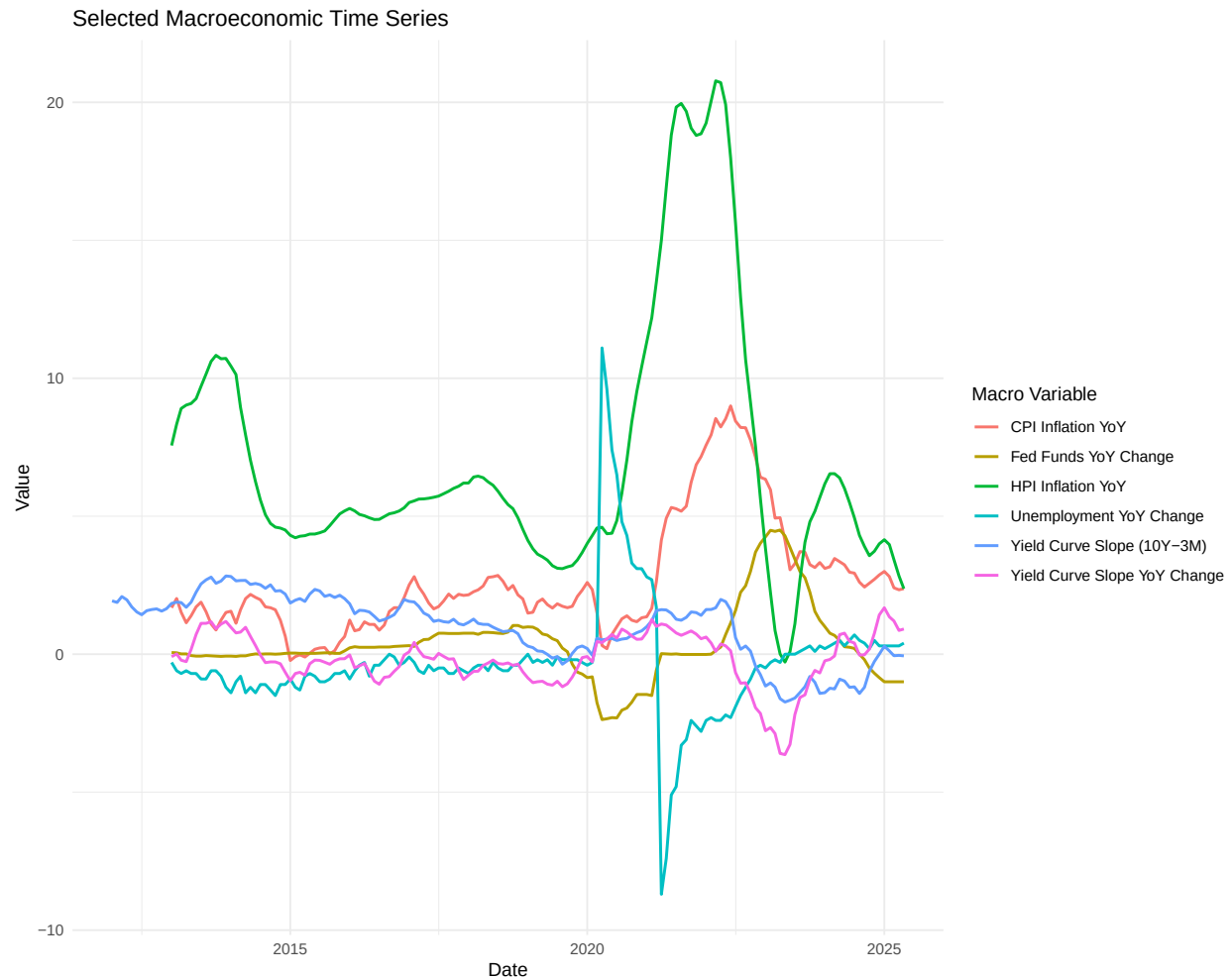
⁶Since “default” was defined as 90+ DPD, this means we removed observations in months $t + 1, t + 2, \dots$ after $Y_{it} = 1$ even if there may be some loans which ended up returning to a non-default state.

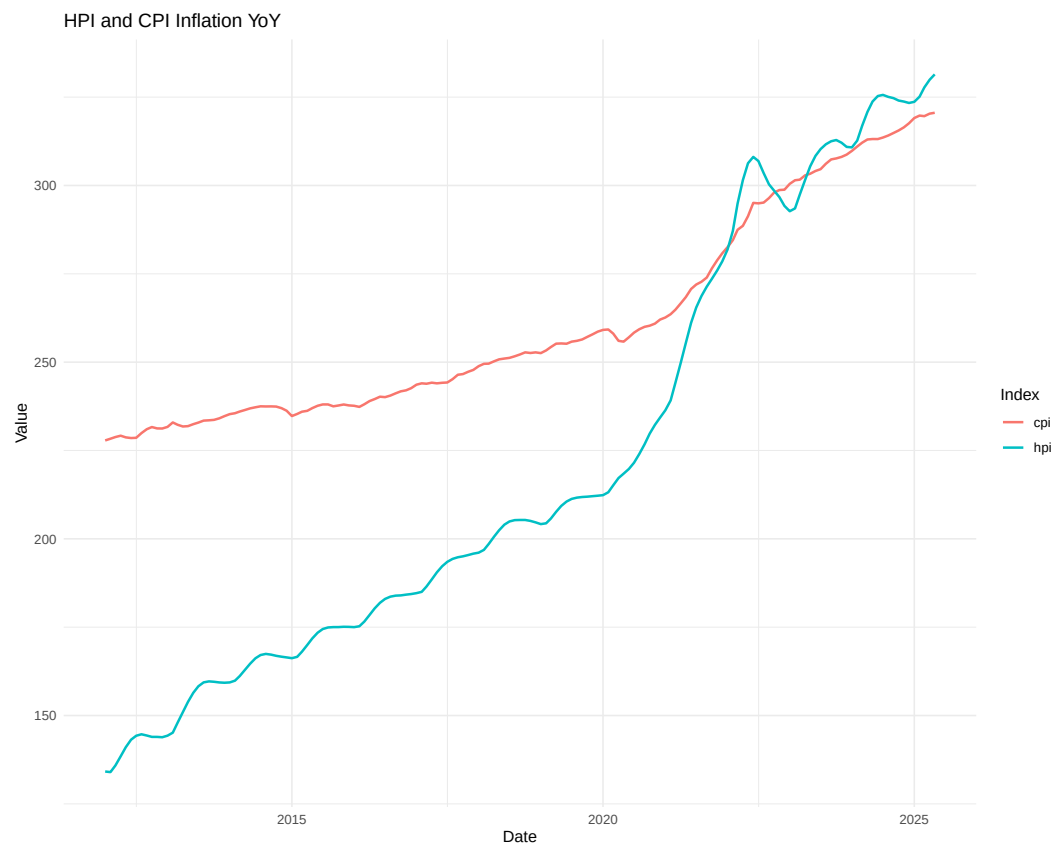
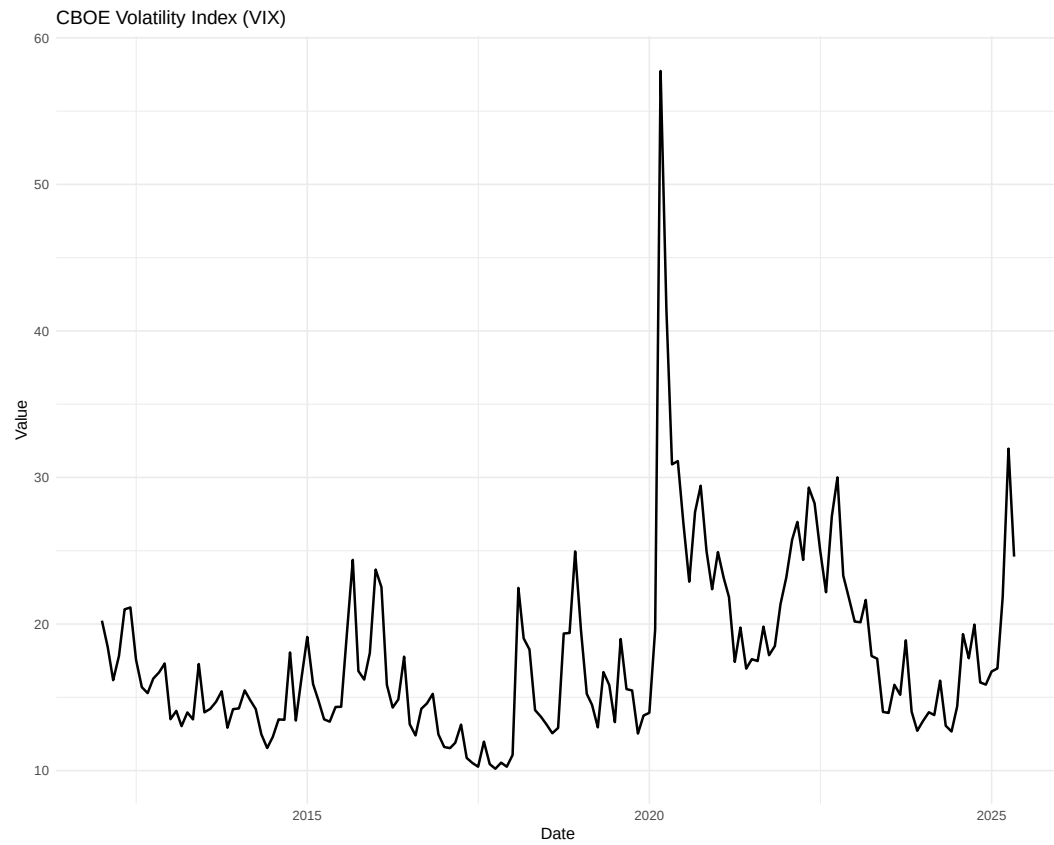
⁷At the final month in our dataset, whether or not a loan defaults next month is unknown.

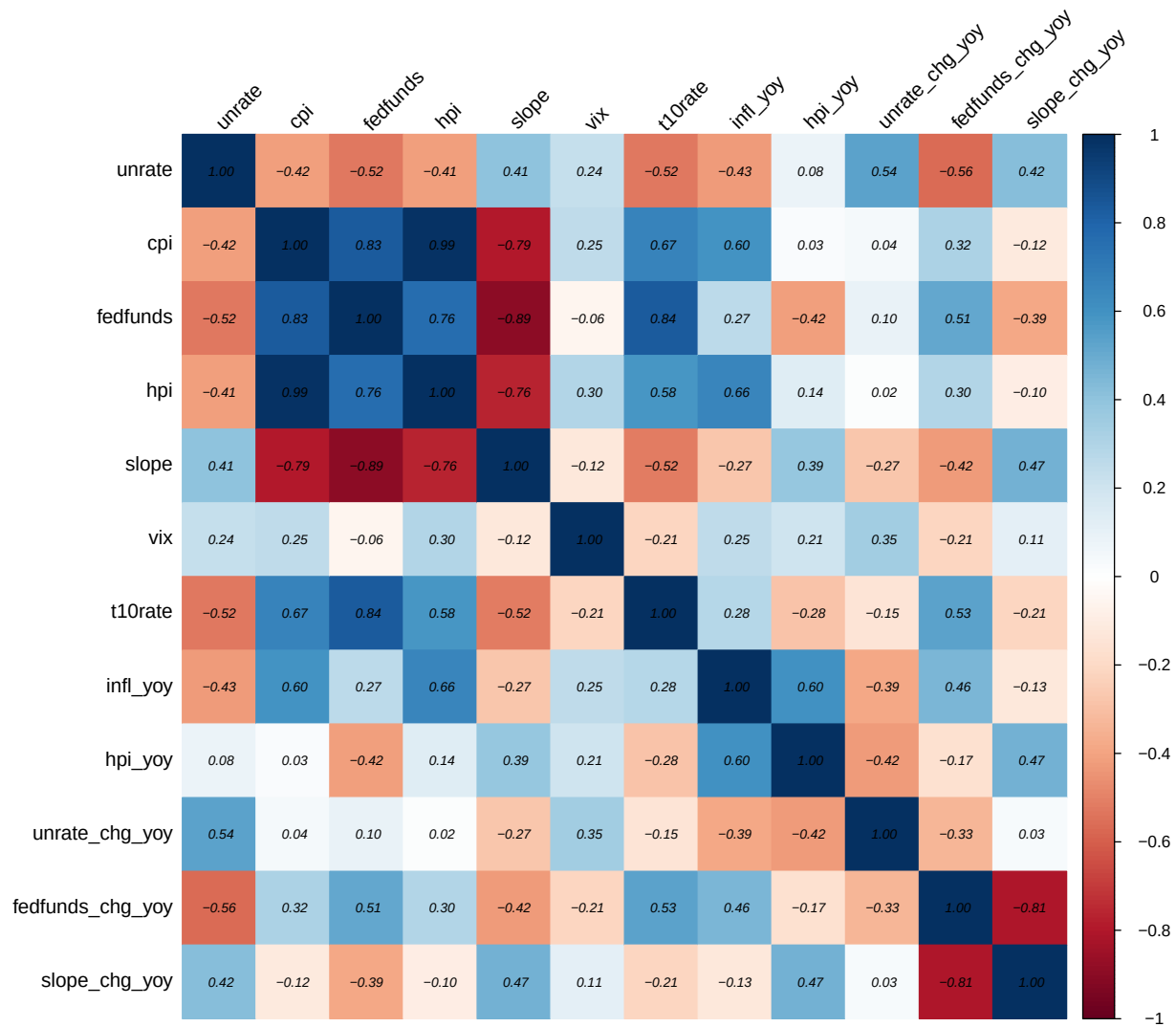
`dti_missing` for `orig_dti` having an NA value, and we create a new variable `orig_dti_imp` which is equal to `orig_dti` when it is not missing and otherwise it is imputed with the median. This way, we can use both `dti_missing` and `orig_dti_imp` as variables in our model.

Lastly, note that all loans in the data happen to be *fixed* rate mortgages (FRMs) and *not* be prepayment penalty mortgages (PPMs). For this reason, we cannot include the indicators for either of these factors in our model, and it also means that our model is implicitly conditioning on the loan being an FRM and a non-PPM.

Macroeconomic Data







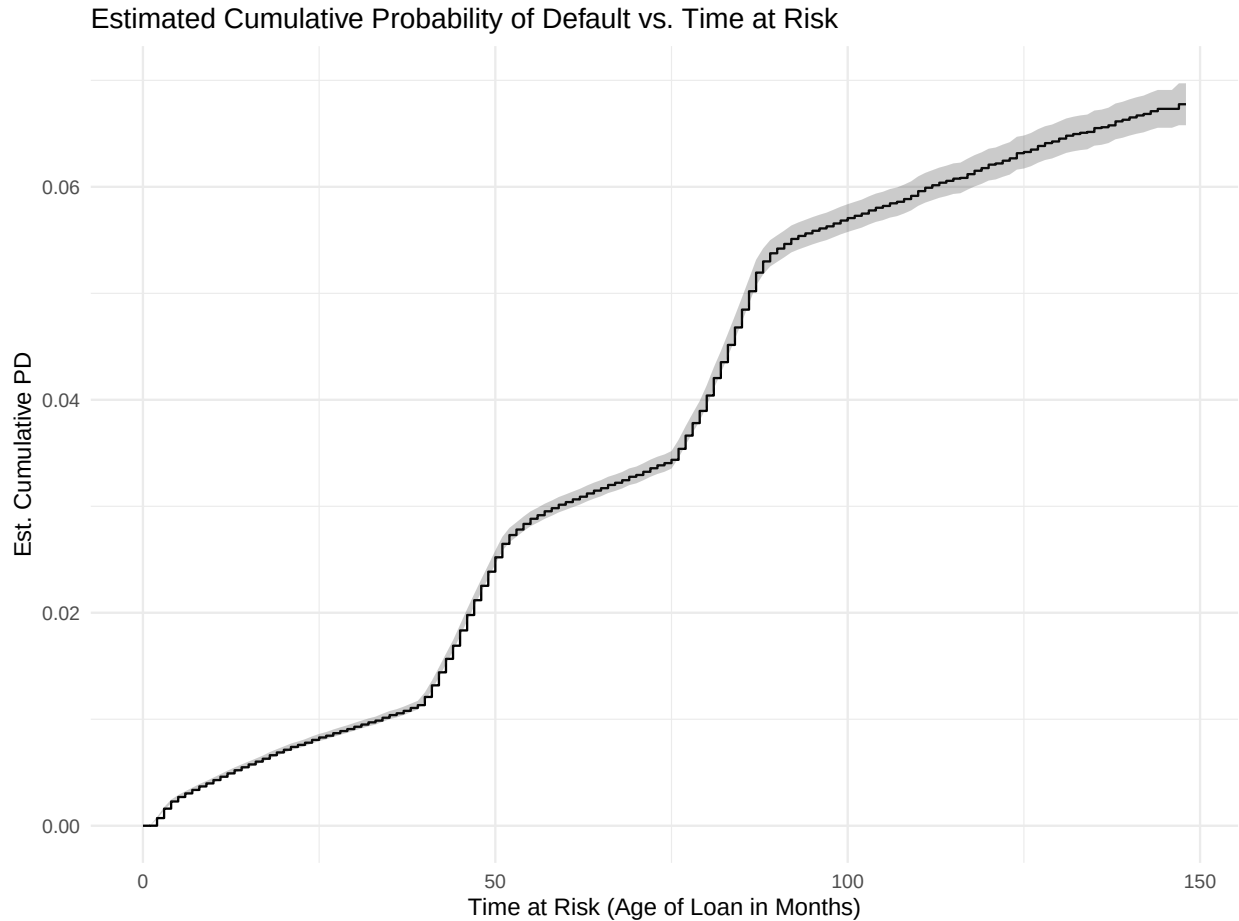
Macroeconomic variables are included in the model as exogenous time-varying covariates, which allows the model to condition on observed (lagged) macroeconomic variables and hence help the model learn the overall economic environment that lenders face at a given point in time.

It is necessary to lag the macroeconomic features by (at least) one month because the data for month $t - 1$ is typically only released during month t .

EDA, Variable Selection and Transformations

Univariate Survival Analysis

The Kaplan-Meier estimate of the cumulative incidence function is provided below. In our context, cumulative incidence is the cumulative PD.



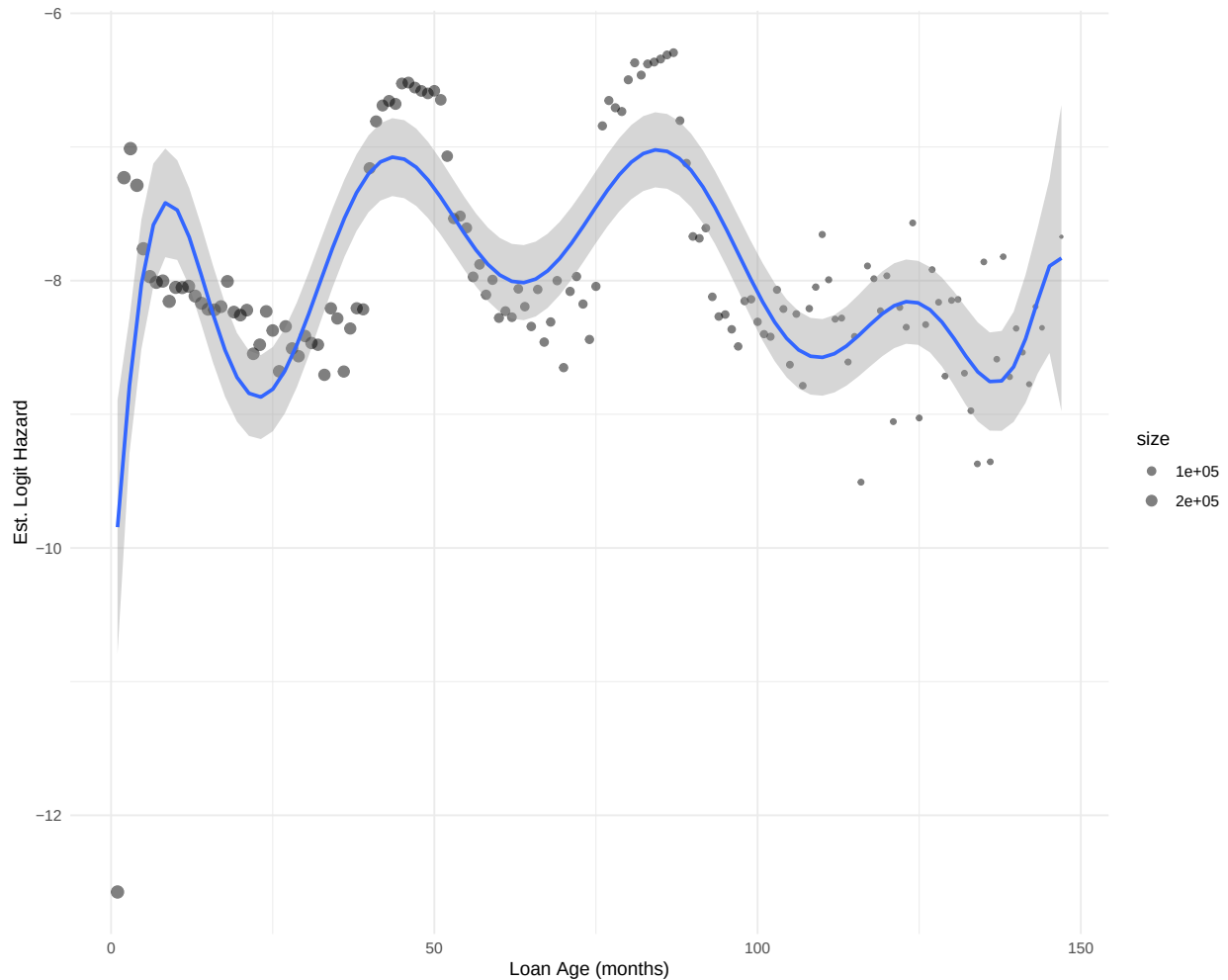
The estimated cumulative PD (eCPD) curve shows spikes at around 40-50 months and 75-90 months, which is suspected to be the effects of macroeconomic shocks from those periods of time (i.e. COVID). For the 2016 origination sample, a loan age of 40-50 months roughly corresponds to the time period of 2020 Q2 on average, and similarly for the 2013 origination sample, a loan age of 75-90 months roughly corresponds to the time period of 2020 Q2 on average. If we look more closely at the very start of the eCPD curve for small loan ages, we see that it is very subtly increasing at a faster pace than the prolonged steady rates of growth we see for later months, which aligns with the fact that the 2020 origination sample was in the calendar time period of 2020 Q1-Q2 for these loan ages.

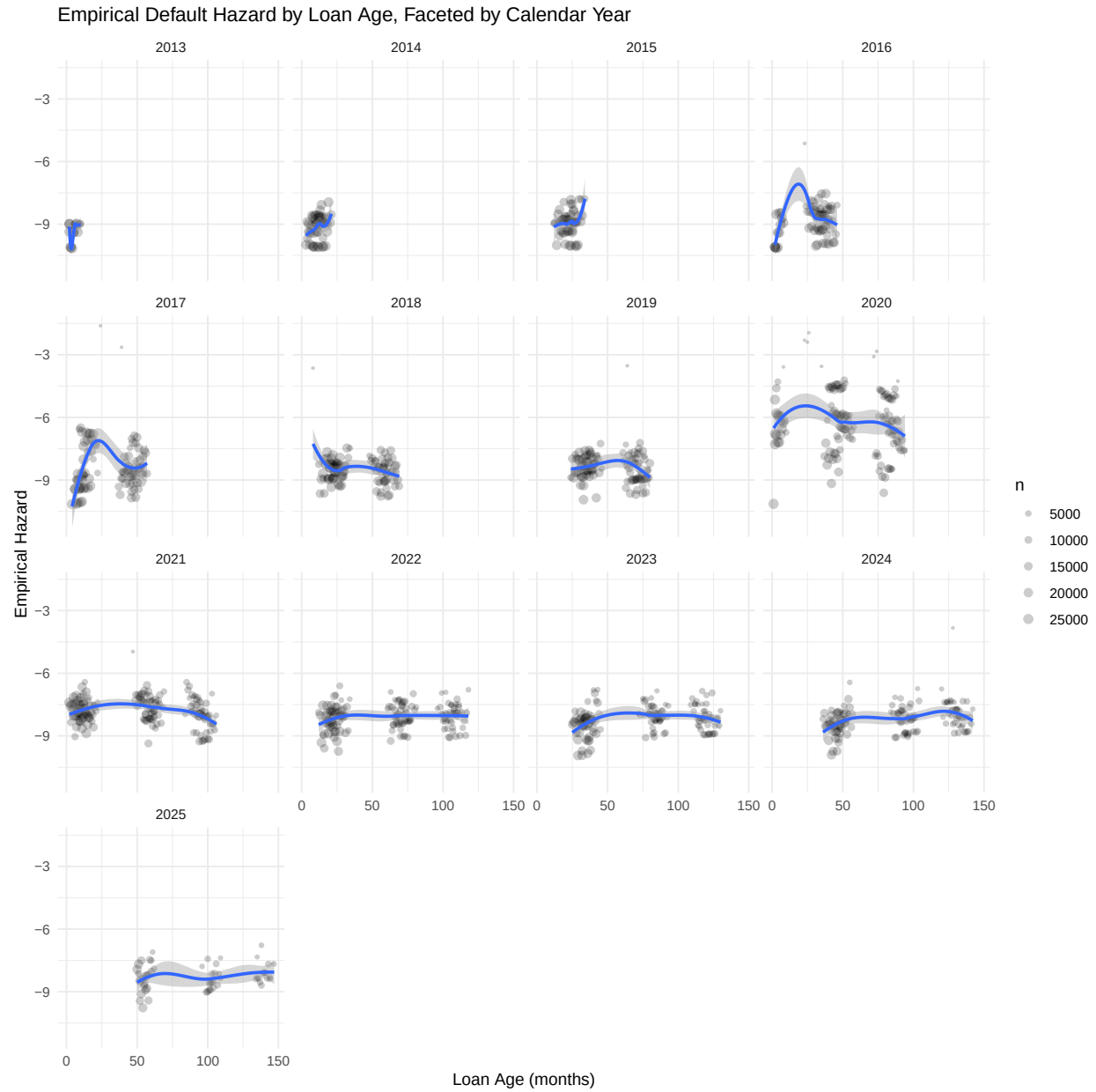
The goal is to use macroeconomic covariates such as the unemployment rate and the house price index in our PD model so that we can try to capture the effects of macro shocks on PD and so that we can identify the possible effect of other variables like loan age without the issue of confounding.

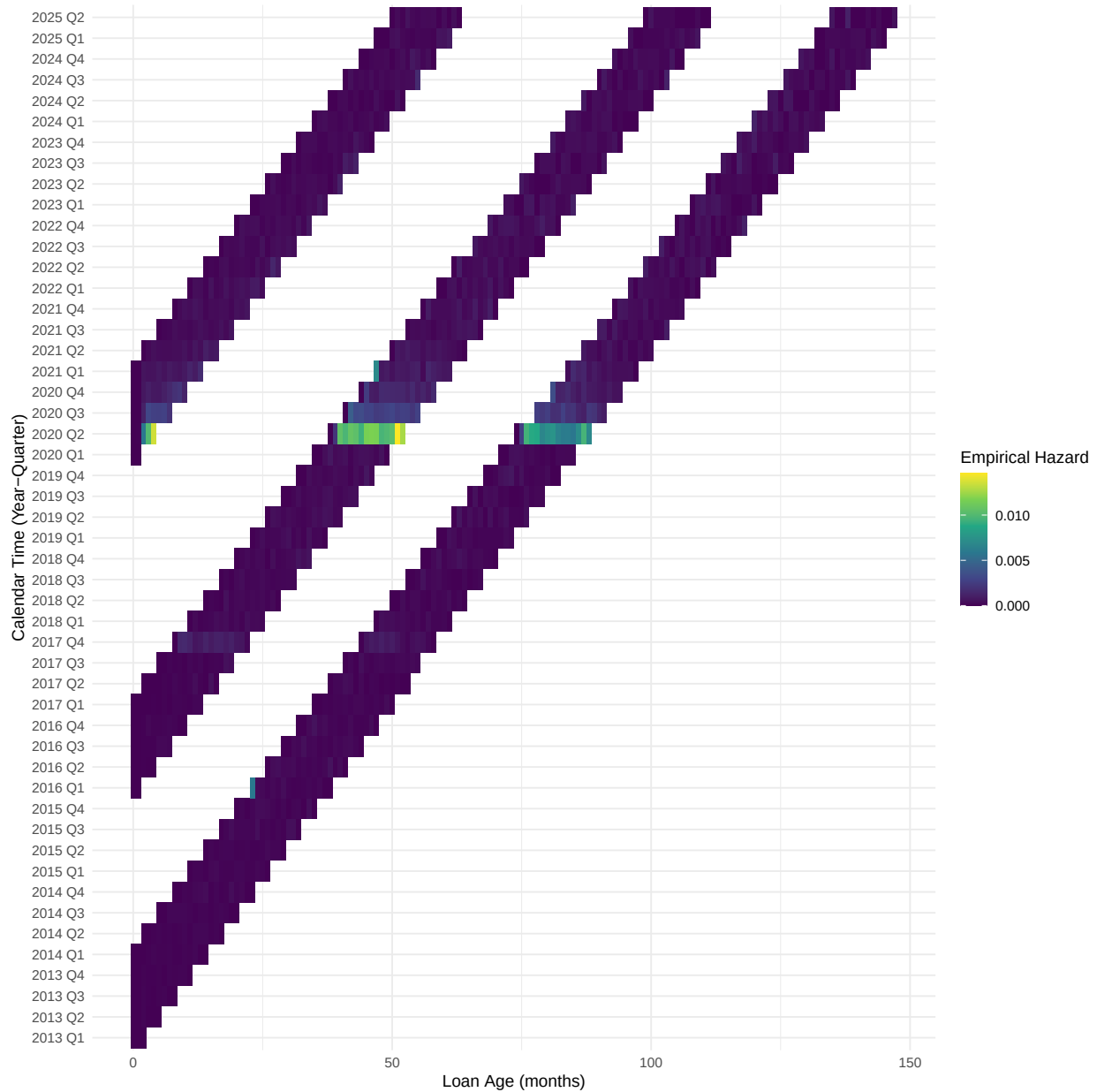
Feature Standardization

Continuous predictors are used in their original units in the current analysis. In production credit risk settings, continuous features are typically standardized during estimation to improve numerical stability and comparability across predictors, with scaling parameters fixed using the training data sample statistics and treated as part of the model. Distributional shift would be monitored (e.g. using the Population Stability Index), with model refitting considered when thresholds are exceeded as opposed to rolling re-standardization.

Identifying Possible Age of Loan Effects



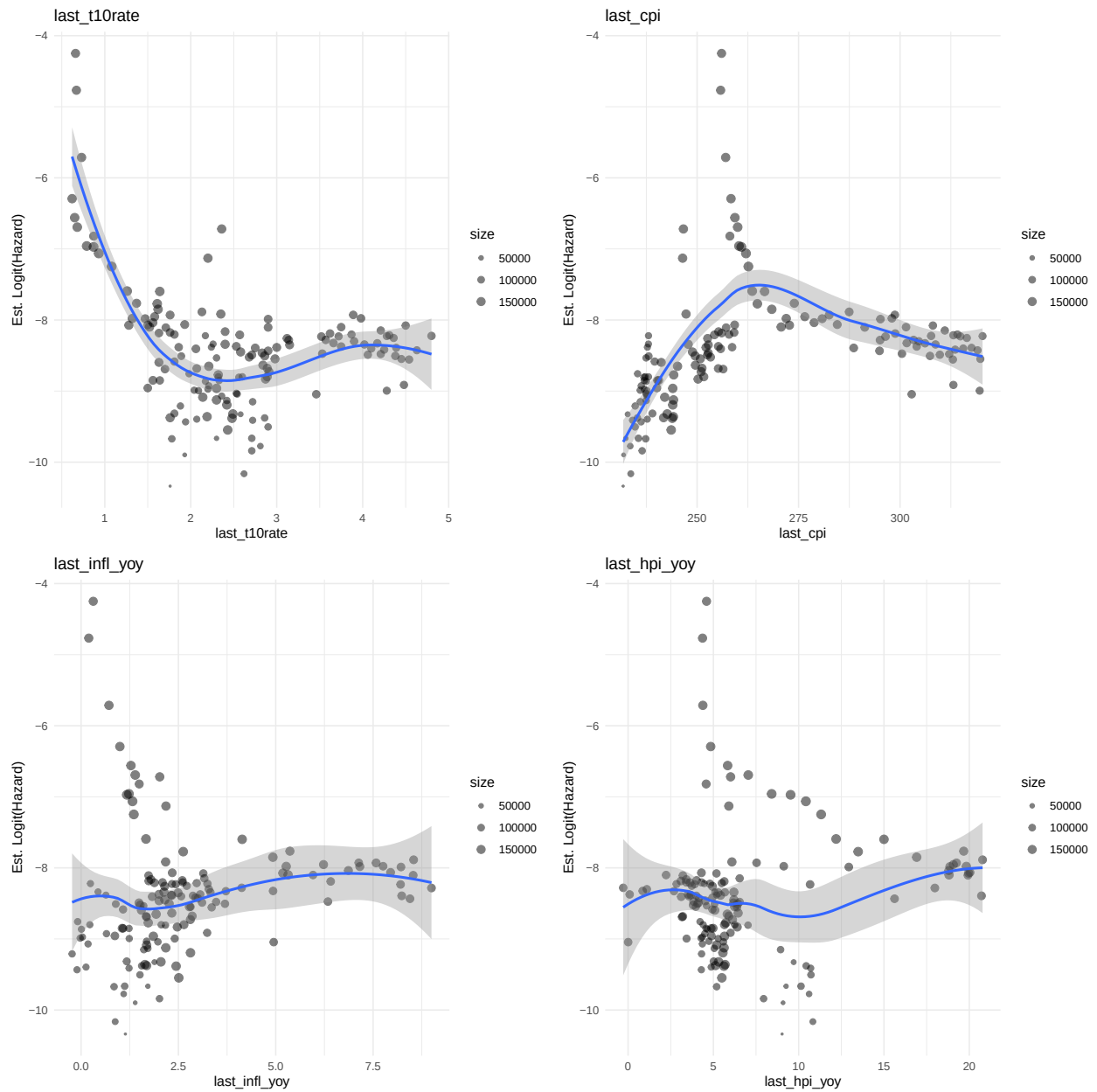


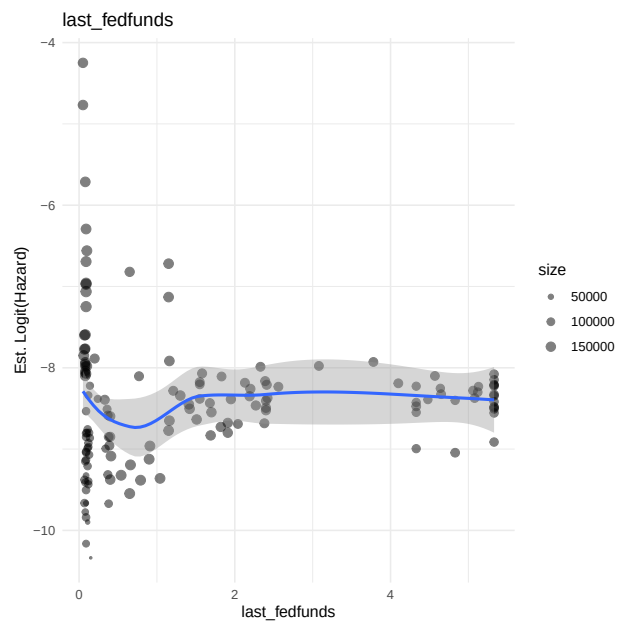
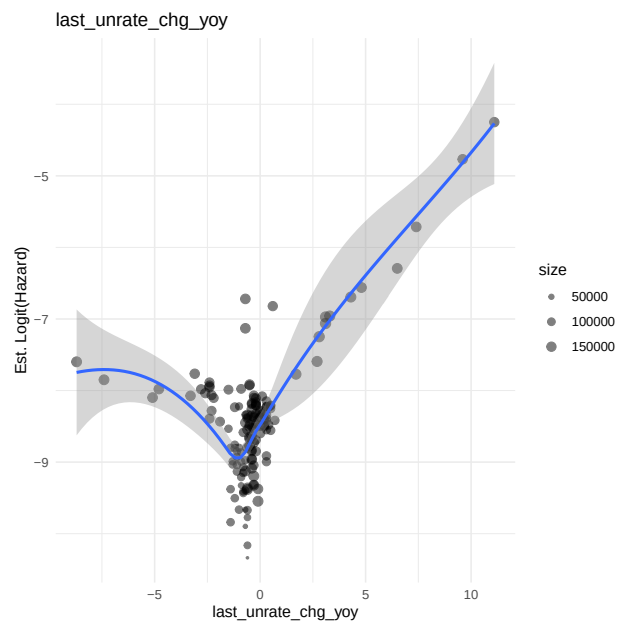
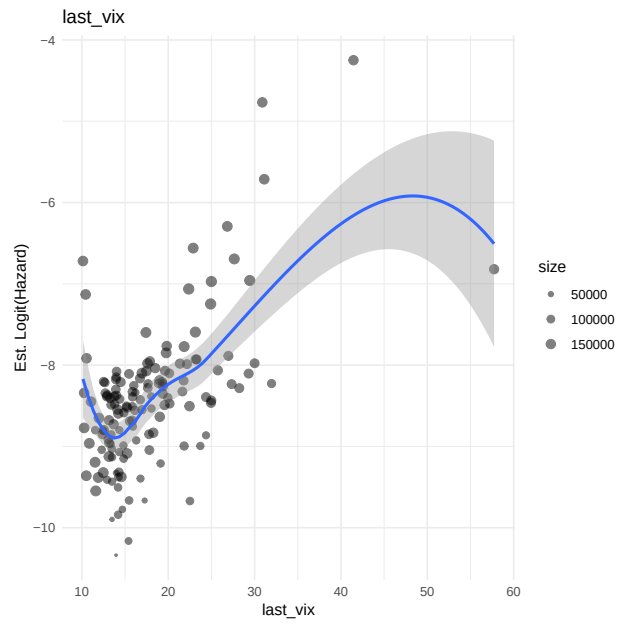
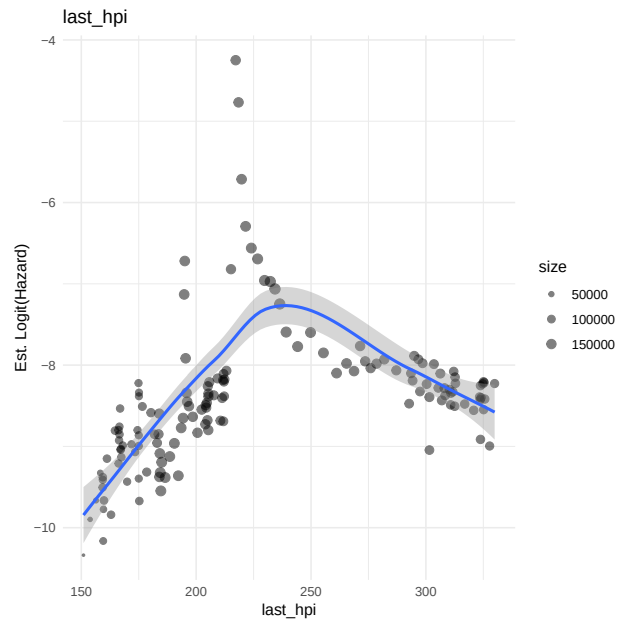


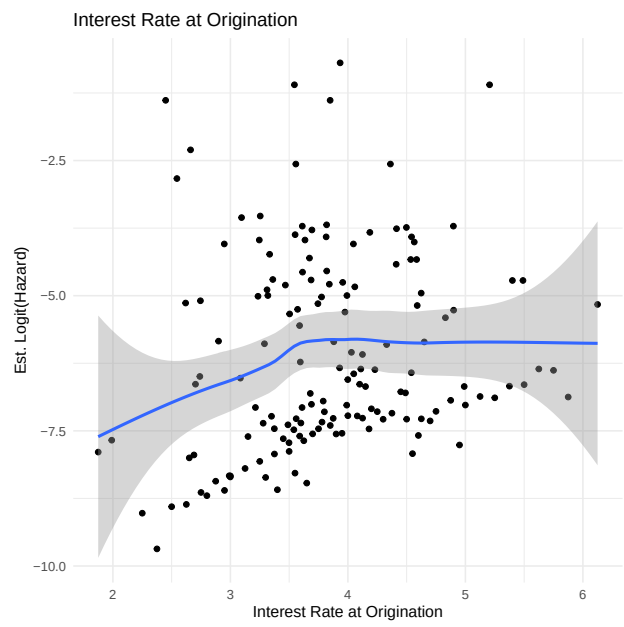
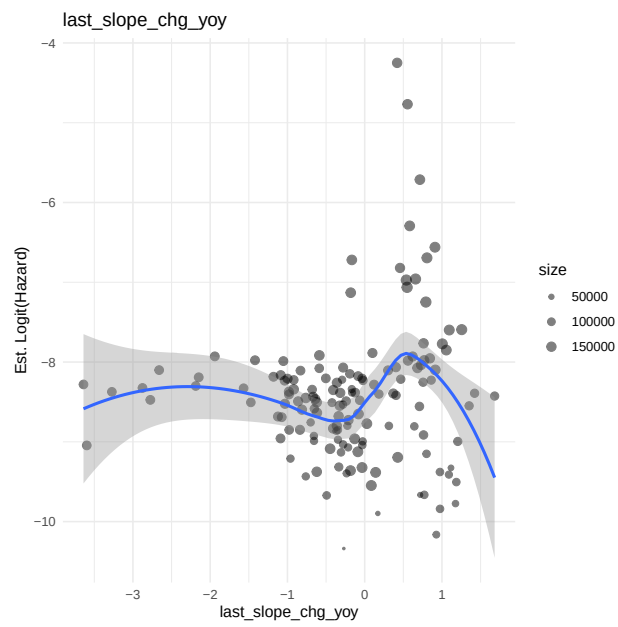
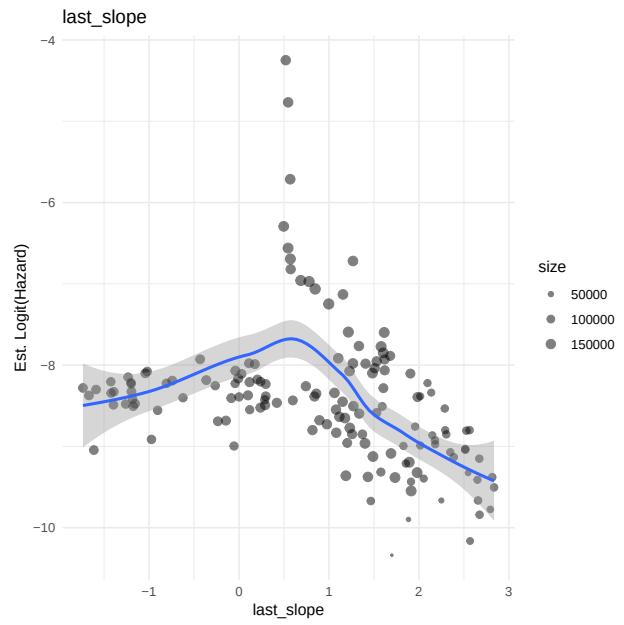
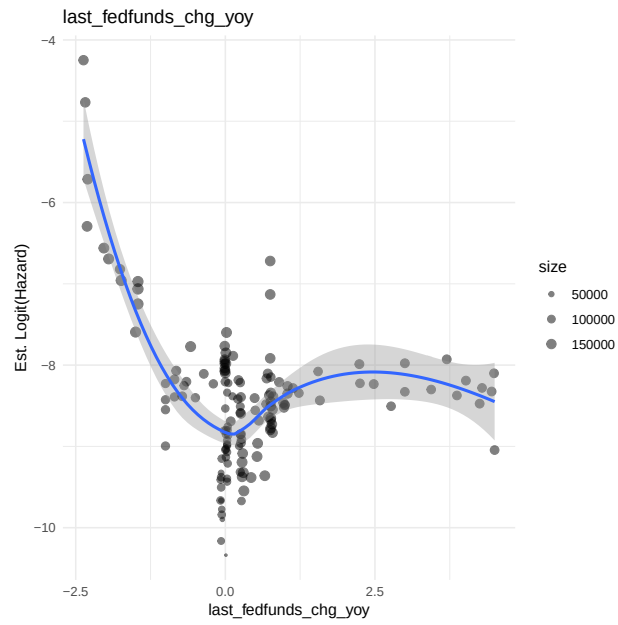
From the heat map above there does not seem to be any association between hazard and loan age. However, we do see clear spikes in hazard across loans of *all ages* during the time period of 2020 Q2. In the year-faceted scatterplot, we observe 3 clusters (one for each origination) for most years, and it is difficult to identify any relationship with loan age within each year.

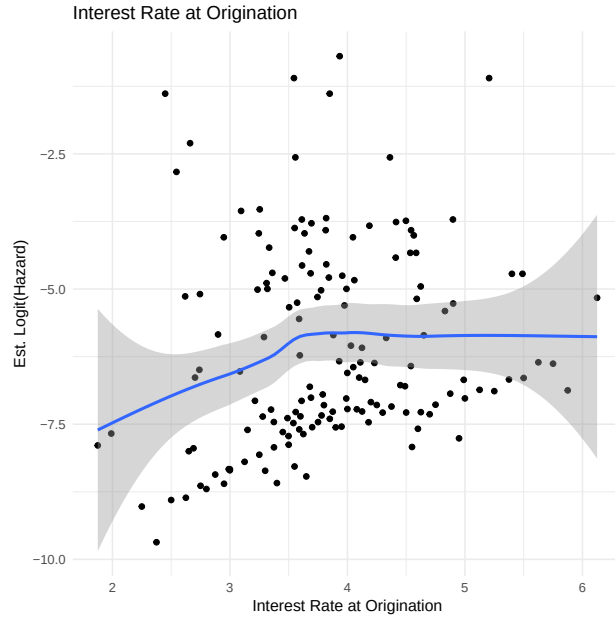
Thus, it seems that macroeconomic conditions play a much more important role in the event of default than age of loan does. However, loan age may still be a useful control in our model since we are using many time-dependent macro covariates. Thus, we will fit models both with and without the age of loan variable and compare them.

Selecting and Transforming Macroeconomic Features









We will use VIX (`last_vix`) as a linear predictor. There is only one point which is a very clear outlier from the overall linear association, which is what makes the LOESS curve appear nonlinear.

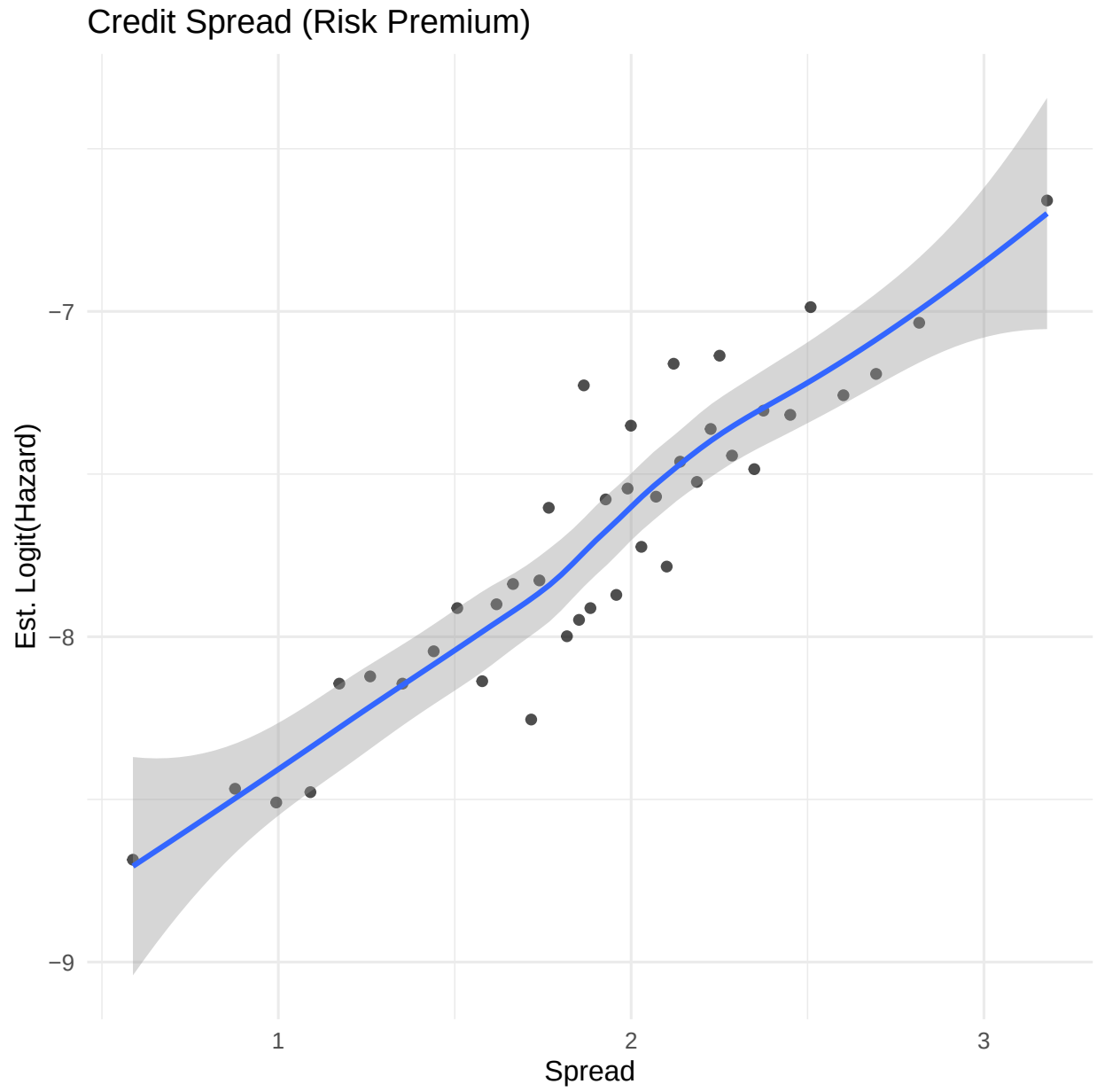
Based on the patterns observed in the plots above and the economic context, we make transformations to many of the other variables in an effort to see if we can get better candidates for linear predictors while maintaining interpretability.

Before moving on to the transformations, we note that there are some variables which we can disqualify right away based on the plots.

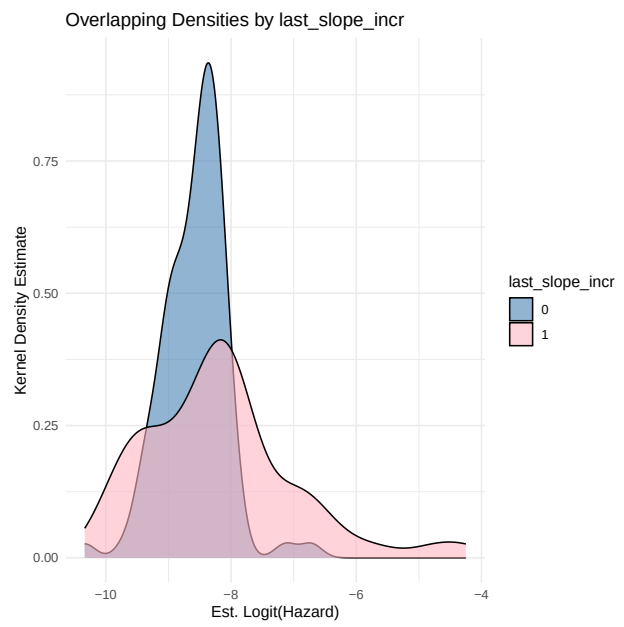
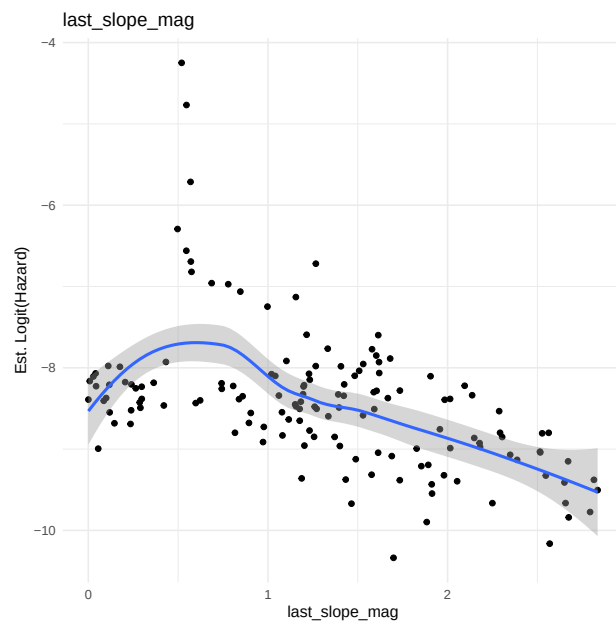
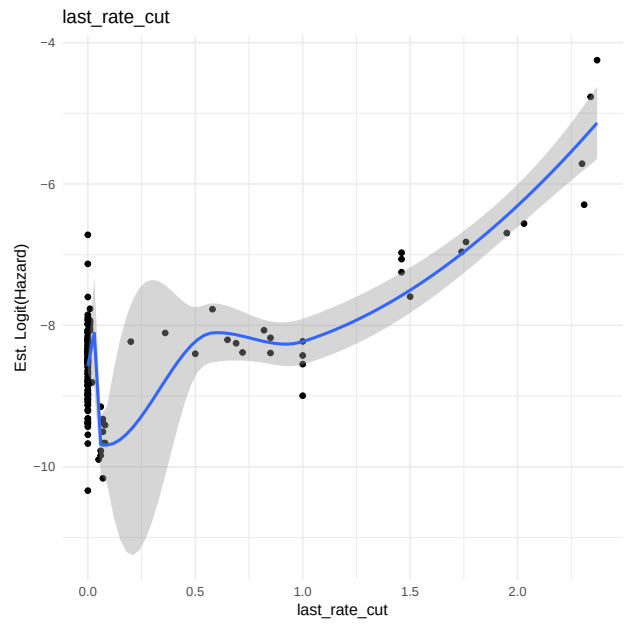
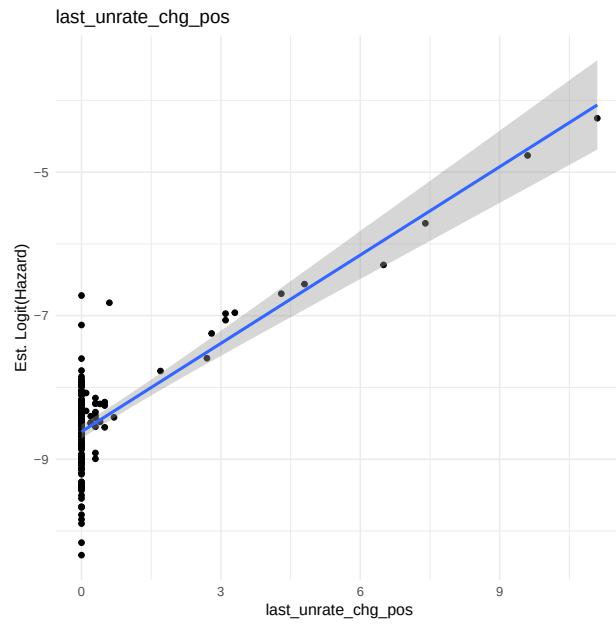
The raw level of HPI (`last_hpi`) and CPI (`last_cpi`) have near perfect positive correlation, and they are both strongly correlated with time, so we suspect that the patterns we observe in the plot are confounded by events in the time periods corresponding to those index levels (i.e. COVID). Hence, we do not use these variables.

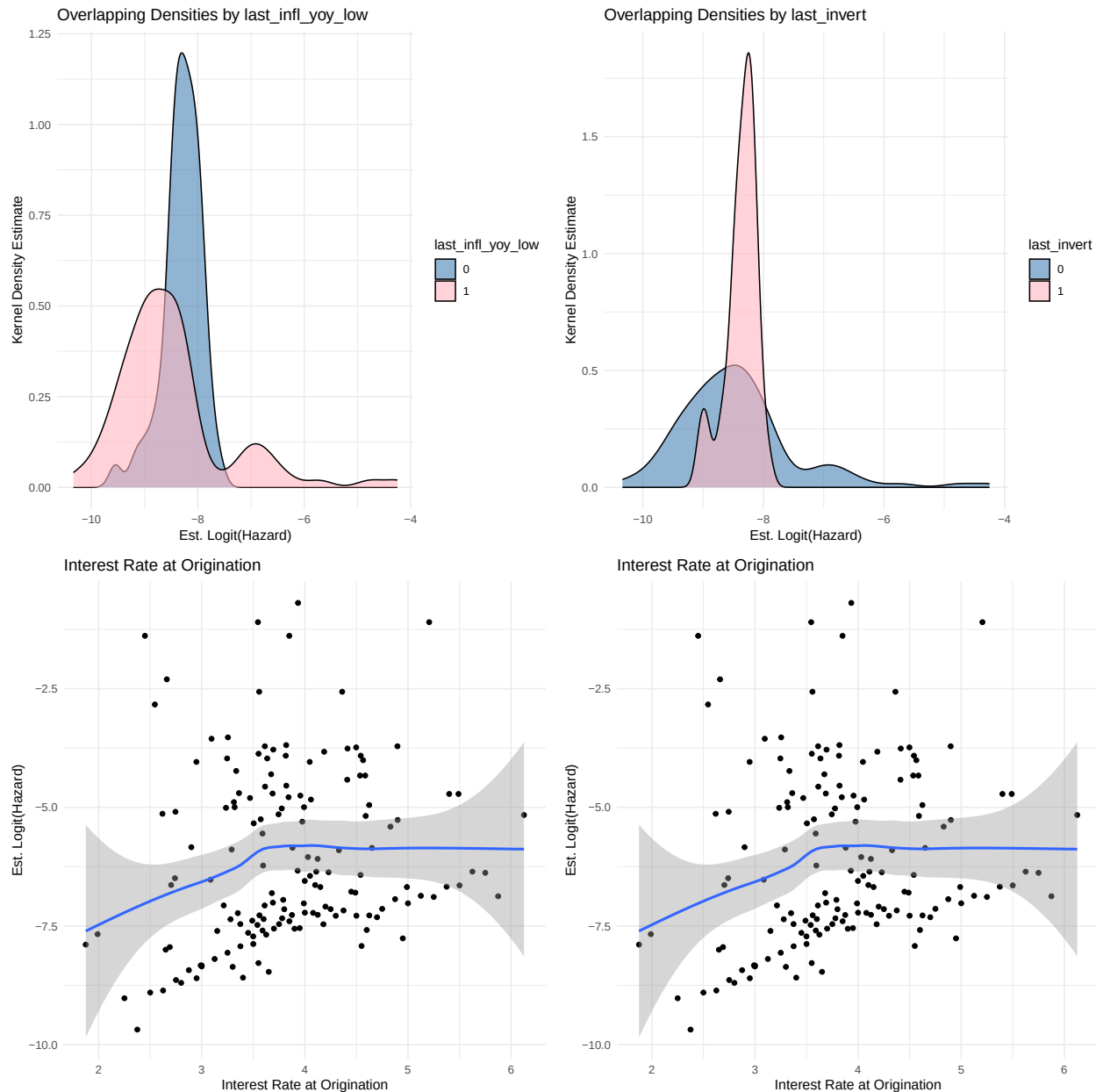
The home inflation (`last_hpi_yoy`), overall inflation (`last_infl_yoy`), and raw level of the federal funds rate (`last_fedfunds`) all show no association with logit hazard, so we do not use them.

We will use the 10-year treasury rate (`last_t10rate`) and the loan rate at origination to create a new variable that should offer both better interpretation and predictive power.



The credit spread shows very clear linear association with the empirical logit hazard.





`last_unrate_chg_pos` has a strong positive linear association with logit hazard. $\text{last_unrate_chg_pos} = \max(\nabla_{12}\text{UR}, 0)$ where UR is the unemployment rate time series.

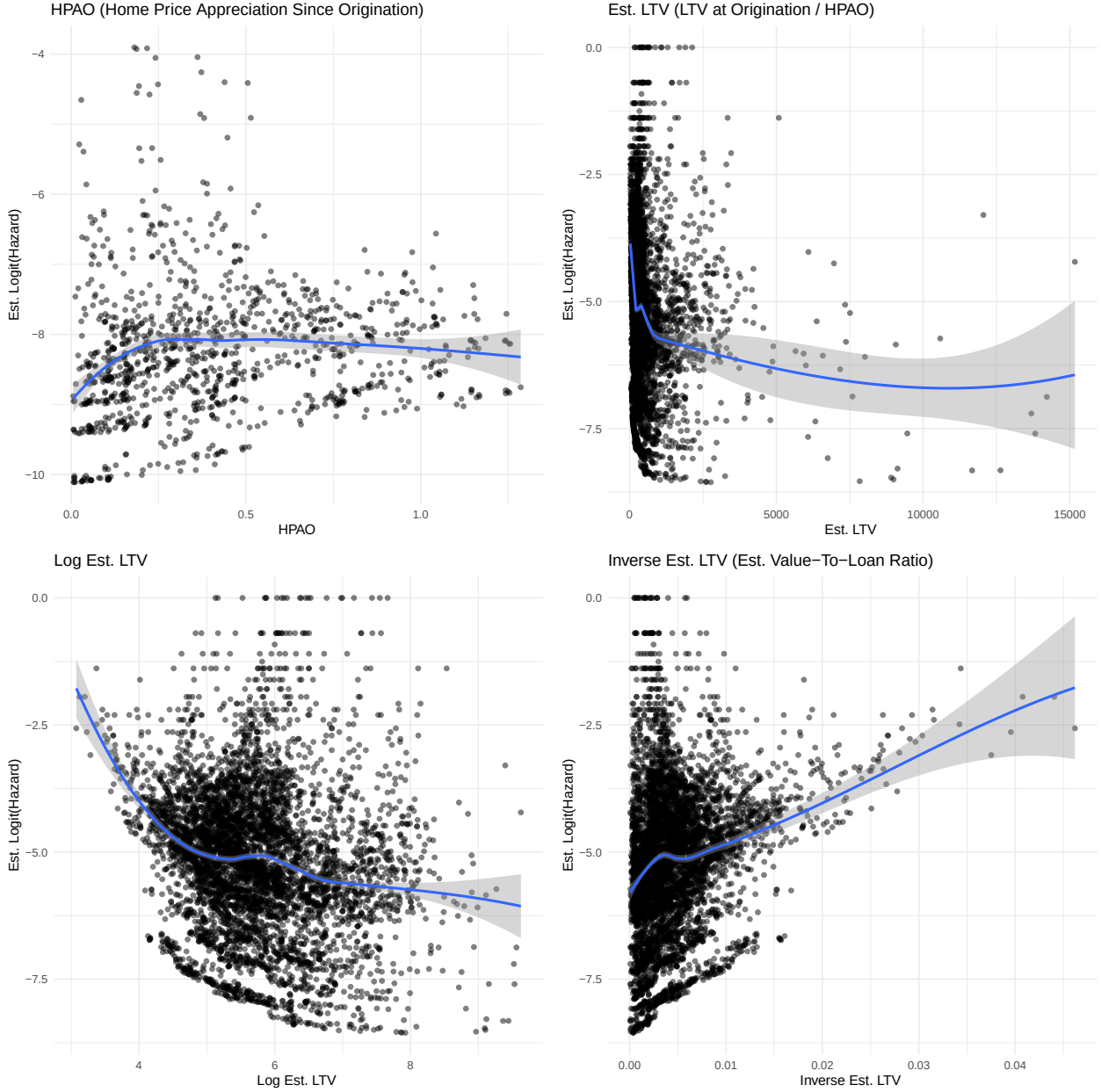
`last_rate_cut` has a moderate-to-strong association with logit hazard that is approximately linear. $\text{last_rate_cut} = \max(-\nabla_{12}\text{FF}, 0)$ where FF is the fed funds rate time series.

The distribution for the empirical logit hazard when `last_infl_yoy_low==1` (last month's YoY inflation was below 2.5%) is shifted left with lower variance, and it is shifted right with higher variance when `last_infl_yoy_low==0`, which suggests a change in distribution, so we use this indicator in the model.

The kernel density estimates for the subset where `last_slope_incr==0` and `last_slope_incr==1` are centered at roughly the same logit hazard value so we do not use `last_slope_incr` as a predictor. For the same reason, we do not use `last_invert` (the indicator for last month's yield curve slope

being negative).

Looking at Approximate LTV



We define the time-varying “Approximate Home Price Appreciation since Origination” (HPAO) variable as

$$\text{HPAO}_i(t) = \text{HPI}(t) / \text{HPI}(t_0^{(i)})$$

where HPI_t is the Home Price Index at time t and $t = t_0^{(i)}$ corresponds to the time of origination for loan i . We mentioned earlier that the LTV (Loan-To-Value) ratio for a given loan i at time $t > t_0^{(i)}$ is estimated by Freddie Mac using their Automated Valuation Model, which is because the value of the home for loan i is typically unknown at an arbitrary time after origination. Using the $\text{HPAO}_i(t)$ variable defined above, we estimate $\text{LTV}_i(t)$ as follows.

Let L_i be the original loan amount for loan i and let $V_i(t)$ be the value⁸

$$\text{LTV}_i(t) = \frac{L_i}{V_i(t)} = \frac{L_i}{V_i(t) \cdot V_i(t_0^{(i)})/V_i(t_0^{(i)})} = \frac{L_i}{V_i(t_0^{(i)})} \cdot \frac{1}{V_i(t)/V_i(t_0^{(i)})} \approx \text{LTV}_i(t_0^{(i)}) / \text{HPAO}_i(t)$$

We use the assumption that the growth in the value of home i is approximately the same as the growth in HPI.

Now we inspect the possible use of the 1-month lagged versions of $\text{HPAO}_i(t)$ and/or $\text{LTV}_i(t_0^{(i)}) / \text{HPAO}_i(t)$ as covariates in the model.

The raw estimated LTV covariate shows a weak nonlinear relationship with logit hazard, and the log transform of the covariate shows a stronger but still somewhat nonlinear relationship with logit hazard. The inverse transform (which is effectively the estimated Value-To-Loan ratio) shows a moderate-strong linear relationship, so this is what we use in the model.

The Value-To-Loan ratio can be viewed as a dimensionless version of the equity of the home:

Let E be the equity of the home, V be the value of the home and L be the loan amount, all at some fixed time. Now we have the following sequence of bijections: $E = V - L \mapsto \frac{V-L}{L} = V/L - 1 \mapsto V/L$

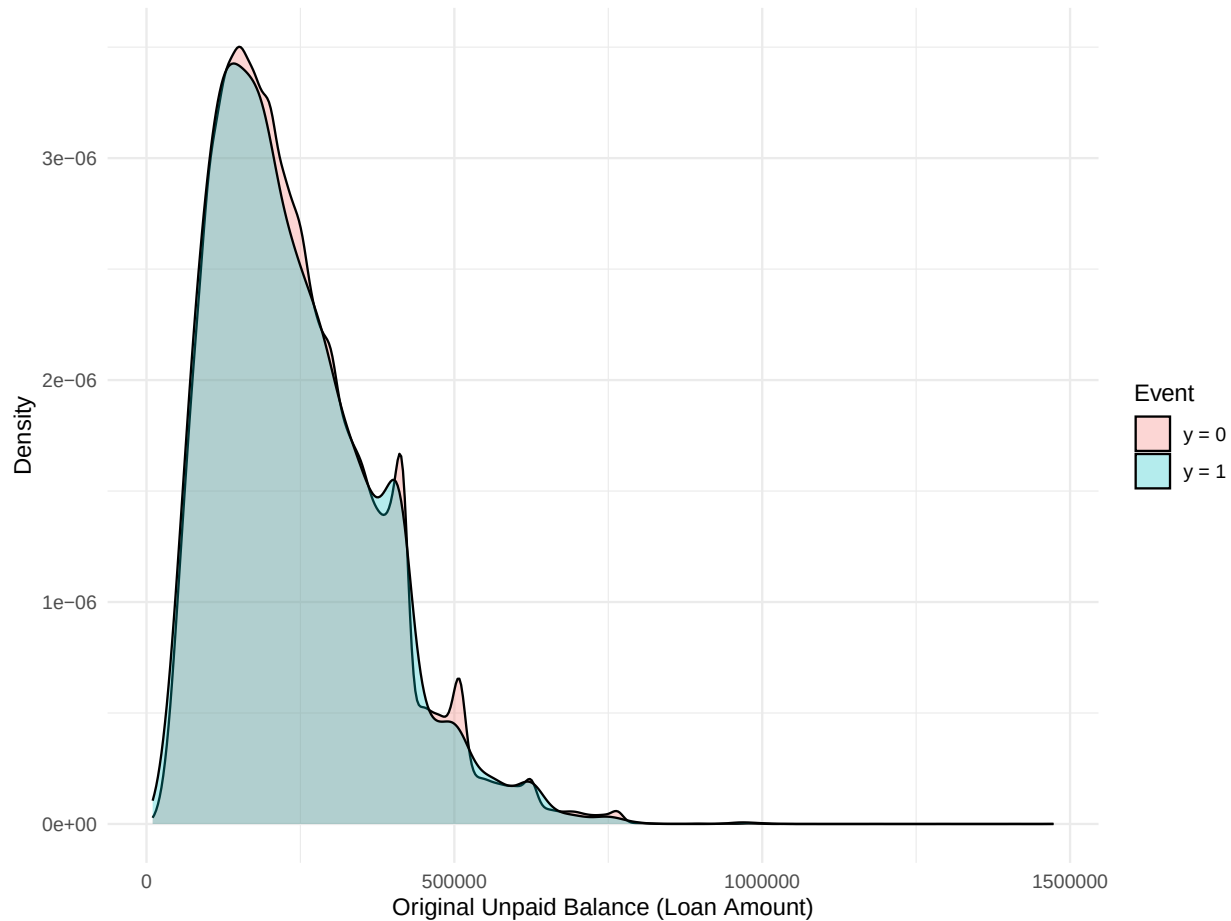
This makes sense economically and aligns with the structural (option-theoretic) view of mortgages, wherein the mortgage is essentially viewed as put option on the home with the strike being the loan amount.

- (Negative Equity Case) $V < L \implies$ default (exercising the put) is economically rational.
- (Positive Equity Case) $V > L \implies$ the put is out of the money and default (exercise) probability is small.

Analyze Rest of Design Matrix

Let's look at the original loan amount as a potential predictor.

⁸When we say "value" here, we really just mean the market price, not necessarily the intrinsic value or fair value in a traditional valuation sense.



The two kernel density estimates are nearly identical so it is clear that there does not seem to be any predictive power in the loan amount variable.

We now look at sample summaries of the outcome across different levels of our categorical predictors to ensure that the event counts are sufficiently balanced across all levels.

variable	level	mu	sd	n
deliq_num	0	0.0000038	0.0019394	18345658
deliq_num	1	0.0006137	0.0247647	94514
deliq_num	2	0.4630041	0.4986294	19786
loan_purpose	N	0.0004298	0.0207264	8809379
loan_purpose	C	0.0005139	0.0226627	3487319
loan_purpose	P	0.0006020	0.0245274	6163260
occupancy_status	S	0.0003404	0.0184466	693317
occupancy_status	P	0.0005075	0.0225218	16274228
occupancy_status	I	0.0005314	0.0230450	1492413
region	North Central	0.0003879	0.0196923	4420807
region	West	0.0004518	0.0212517	5276223
region	South	0.0005917	0.0243169	5991598
region	Northeast	0.0005932	0.0243488	2771330

For `deliq_num` (the number of monthly payments missed), we see that the estimated PD (μ) becomes orders of magnitude higher as the level increases (strong association), so we include it in the model. In fact, the inclusion of this covariate is what effectively makes our model an incomplete Markov transition model, in the sense that it would be complete if we also had models for the 1-month ahead probability of `deliq_num=2` and `deliq_num=1` (which is attainable, just out of scope for this project). Note that we could also remove this covariate to make our model more of a structural economic model rather than a state-dependent model.

We still have class imbalance, as expected.

For each of the other categorical variables, the estimated PD is the same order of magnitude (10^{-4}) across all levels.

For `loan_purpose`, we see that the estimated PD is descending in the order (P,C,N), where P corresponds to purchase borrowers, C corresponds to cash-out refinance (ReFi) borrowers, and N corresponds to no-cash-out ReFi borrowers⁹. The relative growth values in estimated PD as we move across these levels is large, so we include `loan_purpose` in the model.

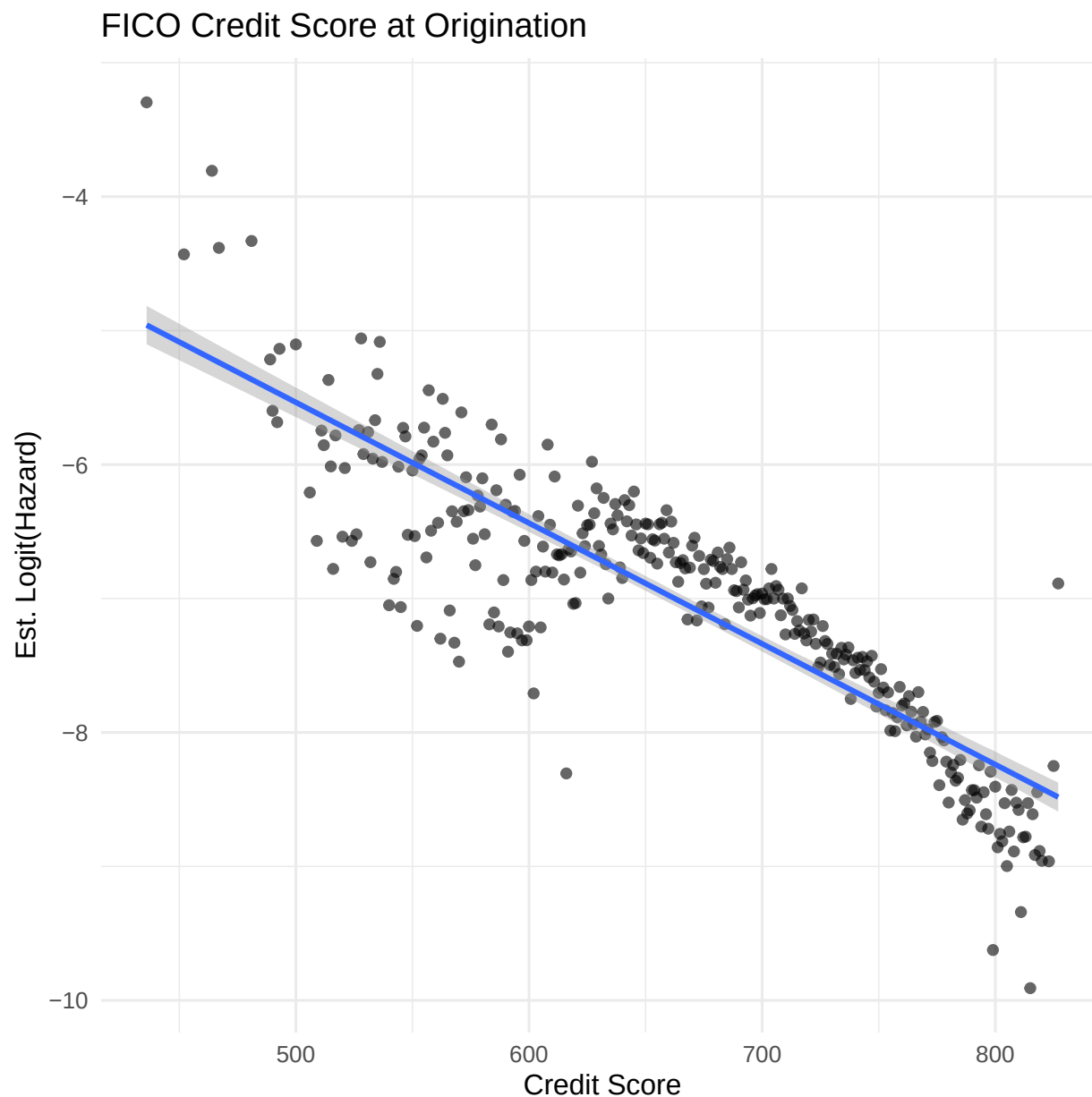
For `occupancy_status`, we see that loans for secondary homes (S) have the lowest est. PD, loans for primary residences (P) have an est. PD that is relatively higher, and investment properties (I) have the highest est. PD although it is only a little higher than that of P. The effect of occupancy status seems weak overall, so we may include it as a weak but interpretable control although it does not seem to be a major risk factor.

For `region`, we observe differences in est. PD across the levels on the same relative scale as seen in the `occupancy_status` factor, so it does not seem to be a major risk factor although once again it may be useful to include in the model as a control, especially since the effects of economic shocks can vary with respect to regional geopolitical status.

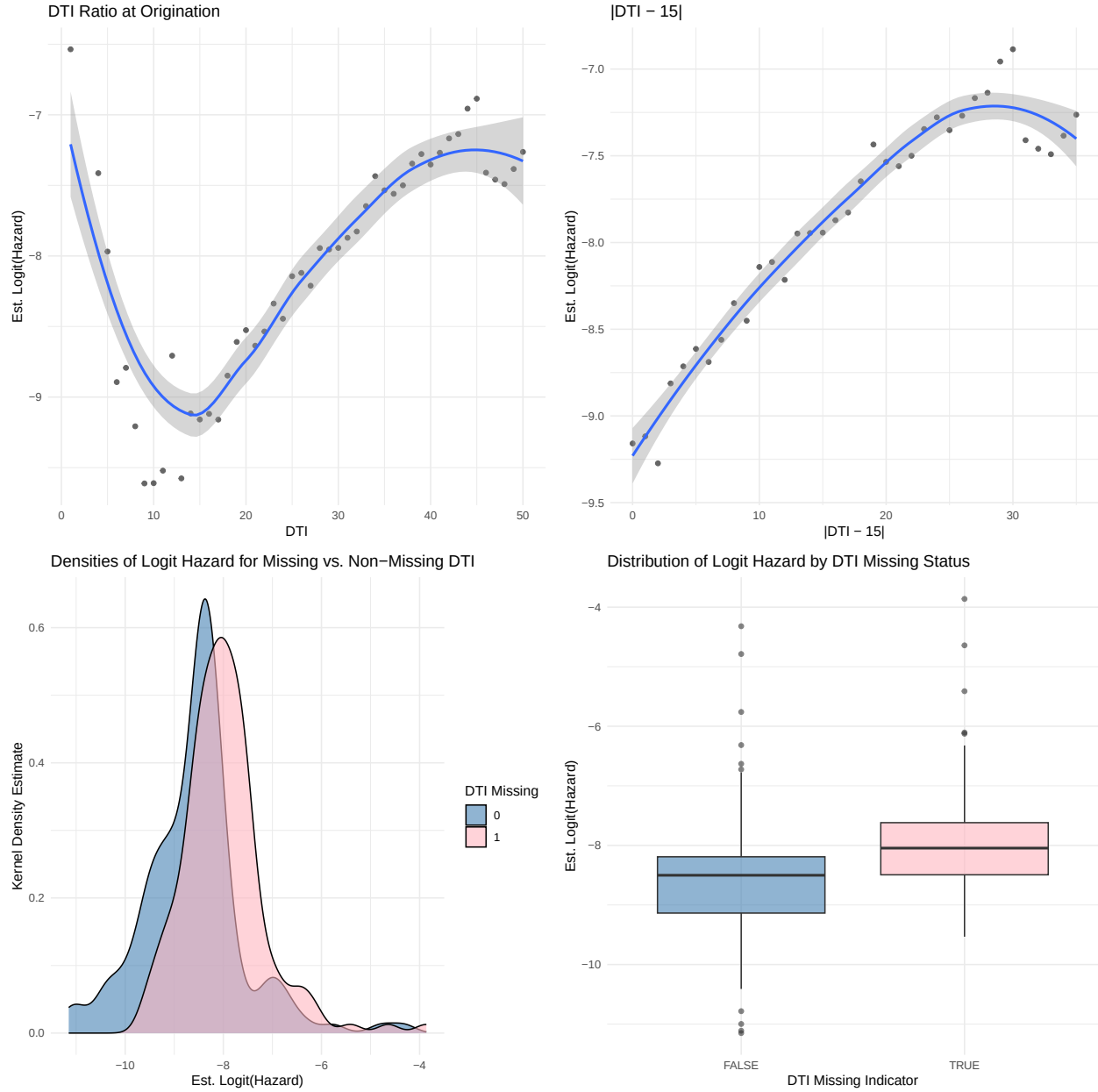


We include the indicator for term of the loan being at least 25 years (`long_term`) as a predictor due to the apparent distribution shift seen above.

⁹Details on what these terms mean can be found on https://www.freddiemac.com/fmac-resources/research/pdf/understand-ser_guide.pdf under the "Origination Data File" table in column position 21.



Credit score shows a strong negative linear association with logit hazard. From the context of our data, we know that FICO scores are typically used when determining the interest rate (at origination), and the interest rate is a function of credit spread and the 10-year treasury rate. Thus, it may seem redundant to include both the credit spread and FICO score at origination and the credit spread at origination, however the correlation is actually quite low in our sample, which we suspect is reflecting the fact that much more than *just* the credit score goes into loan pricing (determining the interest rate to charge). So we include both in the model. We will check VIF scores later to do a final check of multicollinearity.



dti_missing	haz	sd	n
FALSE	0.0004839	0.0219935	15662858
TRUE	0.0006106	0.0247034	2797100

We include `dti_missing` in the model due to the apparent shift in distribution that we observe across the groups.

The imputed DTI at origination variable clearly has a non-monotone relationship with logit hazard, although `transf_dti = |DTI - 15|` has a strong linear relationship except possibly at the right tail where we see some deviation from that pattern.

We will now train some models on a subset of the data. We select the subset by including all loans

with defaults observed and selecting a larger-sized random subset of the other loans (those that did not default). This is done for computational cost savings.

Note: a more thorough approach would be to use cross-validation (CV) on each of the fitted models so that we can use the estimated out-of-sample errors for each model as a basis of comparison. Due to the very large size and panel structure of the data, we would need to manually implement a very efficient CV program that properly handles panel data, so we leave this as an upgrade to complete in a later version of this project. The R packages `data.table` or `Polars`¹⁰ would be useful here, as they would make dataframe computations much faster and more memory efficient than `tibble`, `data.frame`, or pandas `DataFrame` objects in Python.

	df	AIC
symlin_dti	2	112240.0
symquad_dti	3	112122.4

Squared Term Est. Coefficient: -0.001618427

Even though it may seem like the empirical logit hazard is quadratic in `transf_dti`, due to the plot and the fitted coefficient for the squared term being negative, it seems that any fitted quadratic would be concave and hence would approach $-\infty$ as `transf_dti` $\rightarrow \infty$, meaning that the model would return hazard values near 0 for extremely high DTI values¹¹. This behaviour of asymptotically decreasing hazard at very high values of DTI would not be economically plausible.

It would make more sense that the steep growth in logit hazard flattens out or grows at a much slower pace past a threshold such as, say, `transf_dti` = 30 based on the plot. The linear functional form for `transf_dti` does not capture such a pattern, and instead the linear term would make the hazard approach 1 as DTI grows indefinitely, although such behaviour would still make more sense economically.

As seen in the plot, there is only mild flattening behaviour observed and we have relatively much less data in that tail, so we use with the linear form.

Moreover, DTI values greater than 30¹² are near the typical maximum tolerance for mortgages to be underwritten, and in fact Freddie Mac often caps the tolerated DTI at 45 or 50, which means that our observations for DTI values above 45 must be special cases which somehow qualified for the loan based on other more attractive criteria. This would explain why we observe the empirical logit hazard flattening out for DTI values in that range. This is just one sampling bias from the Freddie Mac dataset, and there are likely many more biases. An effort to remove all sampling bias is out of scope for this project, although we would like to remove some of the bias if possible.

Thus, for practical reasons, we use the linear functional form for `transf_dti` even though the quadratic form achieves better AIC.

We do not include the LTV ratio at origination since we are already using the approximate dynamic LTV as derived and analysed earlier.

¹⁰Polars is available in both Python and R.

¹¹Since $\hat{\beta}_2 < 0$, the quadratic dominates for large $z = |\text{DTI} - 15|$, so $\hat{\beta}_0 + \hat{\beta}_1 z + \hat{\beta}_2 z^2 \rightarrow -\infty$ as $z \rightarrow \infty$.

¹²`transf_dti` > 30 $\iff$ DTI > 45

Dealing with Multicollinearity

We will fit the model on a subset of the data (same as used for AIC-based model comparison done earlier). We use this model for VIF calculations and to compare effect sizes for those predictors that are highly correlated (the rest of the model summary output below can be ignored).

```
## # Check for Multicollinearity
##
## Low Correlation
##
##      Term  VIF   VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      loan_age 2.65 [2.65, 2.66]   1.63    0.38    [0.38, 0.38]
##      last_vix 2.65 [2.64, 2.66]   1.63    0.38    [0.38, 0.38]
##      cr_spread 1.61 [1.61, 1.62]   1.27    0.62    [0.62, 0.62]
## last_infl_yoy_low 1.84 [1.84, 1.85]   1.36    0.54    [0.54, 0.54]
##      last_vtl_est 2.80 [2.79, 2.81]   1.67    0.36    [0.36, 0.36]
##      deliq_num 1.06 [1.06, 1.06]   1.02    0.94    [0.94, 0.94]
##      loan_purpose 1.96 [1.96, 1.97]   1.18    0.51    [0.51, 0.51]
## occupancy_status 1.10 [1.10, 1.10]   1.02    0.91    [0.91, 0.91]
##      region 1.09 [1.09, 1.09]   1.01    0.92    [0.91, 0.92]
##      long_term 1.47 [1.47, 1.48]   1.21    0.68    [0.68, 0.68]
##      credit_score 1.23 [1.23, 1.23]   1.11    0.81    [0.81, 0.81]
##      dti_missing 2.04 [2.04, 2.05]   1.43    0.49    [0.49, 0.49]
##      transf_dti 1.07 [1.07, 1.07]   1.04    0.93    [0.93, 0.93]
##
## Moderate Correlation
##
##      Term  VIF   VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
## last_unrate_chg_pos 6.35 [6.33, 6.37]   2.52    0.16    [0.16, 0.16]
##      last_rate_cut 8.52 [8.50, 8.55]   2.92    0.12    [0.12, 0.12]
##
## Call:
## glm(formula = reformulate(predictor_names, response = "y"), family = binomial,
##      data = sample_df)
##
## Coefficients:
##      Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.317e+01 3.715e-01 -35.451 < 2e-16 ***
## loan_age -4.627e-03 1.061e-03 -4.362 1.29e-05 ***
## last_vix 1.309e-02 3.115e-03 4.202 2.65e-05 ***
## cr_spread 4.603e-02 4.520e-02 1.018 0.30850
## last_unrate_chg_pos 9.762e-02 1.249e-02 7.818 5.37e-15 ***
## last_rate_cut 4.167e-01 5.649e-02 7.377 1.62e-13 ***
## last_infl_yoy_low1 -5.178e-01 5.632e-02 -9.194 < 2e-16 ***
## last_vtl_est -8.384e+00 7.727e+00 -1.085 0.27794
## deliq_num1 3.459e+00 1.803e-01 19.183 < 2e-16 ***
## deliq_num2 1.046e+01 1.242e-01 84.172 < 2e-16 ***
## loan_purposeN 7.171e-03 6.338e-02 0.113 0.90992
```

```
## loan_purposeP      -1.718e-01  5.609e-02  -3.062  0.00220 **
## occupancy_statusP -2.574e-01  8.195e-02  -3.141  0.00168 **
## occupancy_statusS -8.259e-02  1.480e-01  -0.558  0.57685
## regionNortheast   4.848e-02  6.311e-02   0.768  0.44241
## regionSouth       2.187e-01  5.233e-02   4.180  2.92e-05 ***
## regionWest        3.809e-02  5.867e-02   0.649  0.51620
## long_term>=25y    5.563e-02  5.908e-02   0.942  0.34634
## credit_score       4.438e-03  3.787e-04  11.719  < 2e-16 ***
## dti_missingTRUE   1.243e-02  6.626e-02   0.188  0.85116
## transf_dti        -3.048e-03  2.857e-03  -1.067  0.28615
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 113321  on 1528667  degrees of freedom
## Residual deviance:  17690  on 1528647  degrees of freedom
## AIC: 17732
##
## Number of Fisher Scoring iterations: 12
```

The only severe multicollinearity is between `last_unrate_chg_pos` and `last_rate_cut`, which makes sense since the Fed tends to cut rates when unemployment is high. The estimated coefficients are positive for both variables, as expected. From the fitted model above, we see that the effect size of `last_unrate_chg_pos` is around 0.1 after adjusting for the roughly 0.2-sized effect of `last_rate_cut`. This tells us that `last_unrate_chg_pos` has a stronger association with hazard, which makes sense since the Fed's policy rate changes are mostly reactionary whereas the unemployment rate is more of a direct cause. Given the larger effect size and smaller standard error, we use `last_unrate_chg_pos` instead of `last_rate_cut`.

Now we will compare the model with and without age of loan, which is correlated with all of the strongly trending macro variables.

	df	AIC
sample_fit	20	17784.84
sample_fit3	19	17794.19

We keep the loan age variable in the model since dropping it (`sample_fit3`) results in about a 10 point increase in AIC.

	df	AIC
sample_fit5	22	17687.41
sample_fit6	23	17692.91
sample_fit4	21	17701.45
sample_fit	20	17784.84

We get a decently sized drop (a difference of about 83) in AIC relative to the linear form by using

natural splines with `df=2` (`sample_fit4`), although the subsequent drops in AIC from increasing the `df` are very small and primarily reflect the model's ability to absorb residual origination and calendar-time variation rather than a stable age-specific pattern. Given the role of loan age as a control for the baseline hazard rather than a structural driver of risk, we use the low-dimensional spline form with `df=2`. This argument is reinforced by the plots from the Identifying Possible Age of Loan Effects section.

Interactions

Certain interaction terms are effectively already encoded as transformed variables (e.g. `last_vtl_est` is a function of the product $LTV \times HPAO$). We will try a few economically motivated interaction terms and see if they sufficiently improve AIC on our sub-sample of data.

We consider the interaction between loan purpose and occupancy status.

	df	AIC
<code>sample_fit</code>	21	17701.45
<code>sample_fit8</code>	25	17706.07

Based on the small drop in AIC (a difference of about 5), we do not include this interaction term.

Now we try a VIX x credit spread interaction term.

	df	AIC
<code>sample_fit</code>	21	17701.45
<code>sample_fit9</code>	22	17703.19

Again, we see only a very small improvement in AIC with the interaction term, so we do not use it.

Train and Interpret Final Model

Training

```
##
## Call:
## glm(formula = reformulate(predictor_names, response = "y"), family = binomial,
##      data = train_df, model = FALSE, x = FALSE, y = FALSE)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.508e+01  3.484e-01 -43.285  < 2e-16 ***
## last_vix       3.004e-02  2.616e-03  11.484  < 2e-16 ***
## cr_spread      1.808e-01  4.325e-02   4.181 2.90e-05 ***
## last_unrate_chg_pos 1.210e-01  6.504e-03  18.606  < 2e-16 ***
## last_infl_yoy_low1 -1.943e-01  6.325e-02  -3.072 0.002126 **
## last_vtl_est    -4.552e+01  9.728e+00  -4.679 2.88e-06 ***
```

```

## deliq_num1          4.969e+00  2.097e-01  23.689 < 2e-16 ***
## deliq_num2          1.209e+01  1.471e-01  82.193 < 2e-16 ***
## loan_purposeN         -2.865e-02  5.615e-02  -0.510 0.609943
## loan_purposeP         -7.740e-02  5.088e-02  -1.521 0.128223
## occupancy_statusP   5.014e-02  6.612e-02   0.758 0.448242
## occupancy_statusS   6.050e-03  1.263e-01   0.048 0.961805
## regionNortheast     8.119e-02  5.745e-02   1.413 0.157544
## regionSouth         1.742e-01  4.824e-02   3.610 0.000306 ***
## regionWest          -7.795e-02  5.320e-02  -1.465 0.142881
## long_term>=25y      4.859e-02  5.344e-02   0.909 0.363173
## credit_score        2.524e-03  3.378e-04   7.472 7.88e-14 ***
## dti_missingTRUE     -1.492e-02  5.827e-02  -0.256 0.797921
## transf_dti          4.081e-03  2.561e-03   1.593 0.111052
## ns(loan_age, df = 2)1 -7.547e-01  1.297e-01  -5.819 5.91e-09 ***
## ns(loan_age, df = 2)2  1.770e-01  1.143e-01   1.549 0.121497
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 130417  on 12727326  degrees of freedom
## Residual deviance:  20992  on 12727306  degrees of freedom
## AIC: 21034
##
## Number of Fisher Scoring iterations: 15

```

The final variables we used (in the order of the output above) were:

- the lagged VIX level,
- the credit spread of the loan at origination (in percentage points),
- the positive part of the lagged change in the unemployment rate (in percentage points),
- the indicator for the lagged YoY inflation rate (in percentage points) being "low" (at or below 2.5%),
- the lagged estimate for the value to loan ratio,
- the current delinquency state (number of missed payments),
- the purpose of the mortgage loan (e.g. purchase or investment),
- the occupancy status of the property (e.g. primary or secondary residence),
- the region of the property (e.g. North or South),
- the indicator for the term of the loan being "long" (at least 25 years),
- the FICO credit score of the borrower,
- the indicator for the DTI of the borrower not being available,
- the absolute distance of the borrower's DTI from a level of 15 (with median-imputed DTI), and

- the age of the loan (in months).

Interpretation

We will now provide some examples of parameter interpretation for this model. Recall from the data background section that all of the mortgages in our final cleaned dataset were fixed-rate mortgages without prepayment penalties. Thus, from here on, when we talk about “loans” we specifically mean fixed-rate mortgage without prepayment penalties.

For any fixed k , we have

$$\begin{aligned}\beta_j &= \text{logit } \pi_{it}|_{x_{itj}=k+1} - \text{logit } \pi_{it}|_{x_{itj}=k} \\ &= \ln \left(\frac{\pi_{it}/(1 - \pi_{it})|_{x_{itj}=k+1}}{\pi_{it}/(1 - \pi_{it})|_{x_{itj}=k}} \right)\end{aligned}$$

where j is the index for a continuous predictor x_{itj} and β_j is the corresponding component of the loan-level $\beta^{(L)}$ or macro-level $\beta^{(M)}$ parameter vector.

Our model based estimate for the odds ratio (OR) of the event of default next month for one additional percentage of credit spread vs. some fixed credit spread is 1.198 while adjusting for the selected macroeconomic variables and loan-specific variables.

This can be rephrased; we estimate that a 1% increase in credit spread is associated with a 19.8% relative growth in the odds of default next month while adjusting for the selected macroeconomic variables and loan-specific variables.

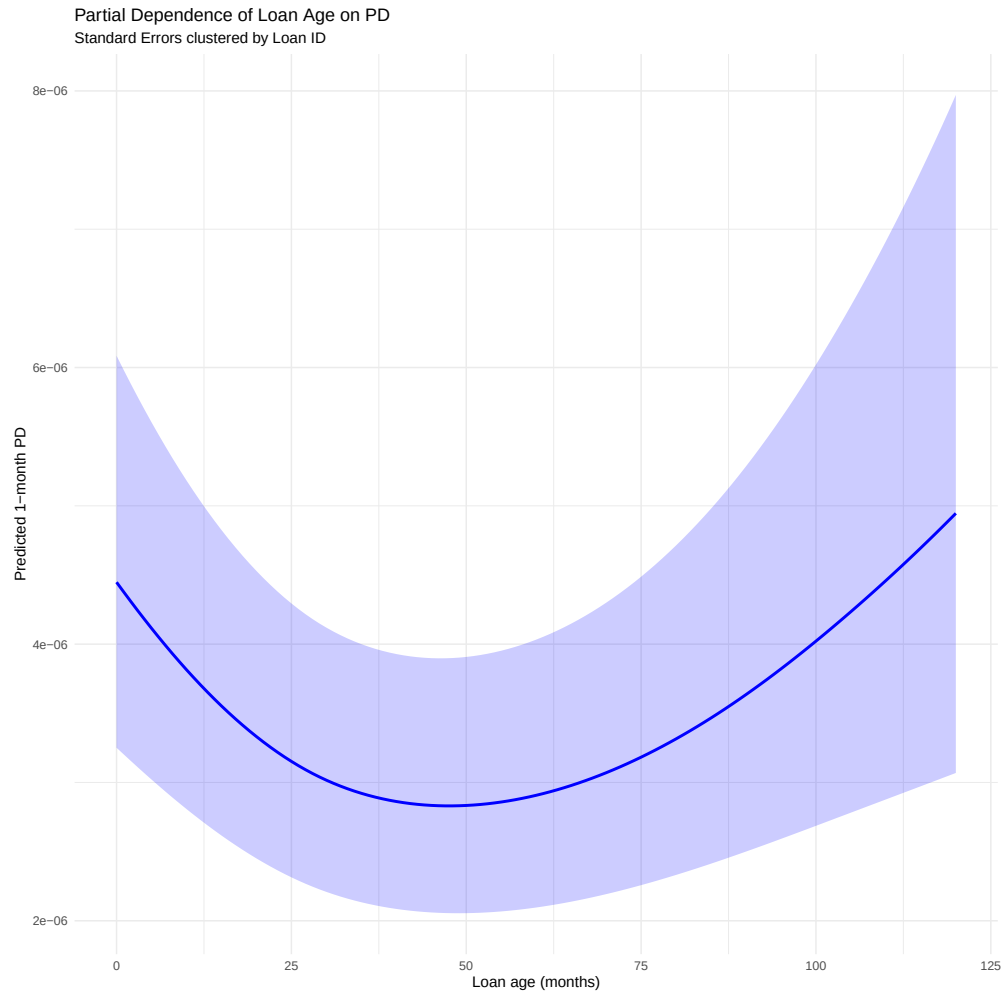
We estimate that the OR of default next month for a loan that has currently missed one payment vs. one that has not missed any payments is 143.823 adjusted for the selected macroeconomic variables and loan-specific variables.

Now we look at the predicted PD values as the number of missed payments increases while holding the other predictor values constant as some manually chosen values; we select somewhat average values for the loan-level covariates (purchased property, primary residence, modest credit score of 750, etc.) although we choose a decently elevated VIX value of 17 and a high credit spread of 3%.

deliq	pd	rel_increase
0	2.22045e-16	NA
1	1.10168e-11	49614.431
2	1.36417e-08	1237.262

The estimated 1-month ahead probability of default for this loan under these macro conditions increases several orders of magnitude as the number of currently missed payments increases.

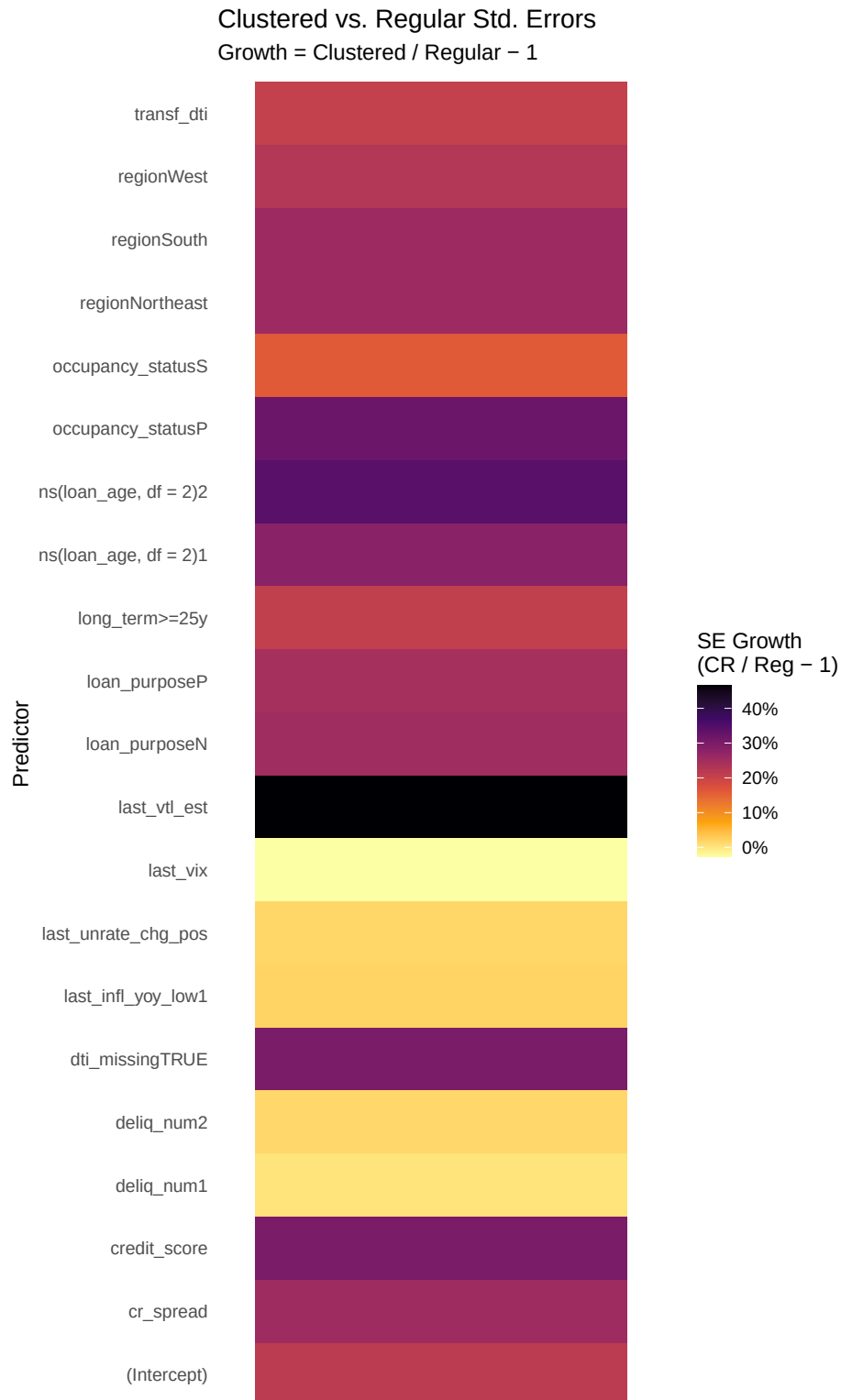
Since we used splines for the loan age predictor it is less easily interpretable, so we create a partial dependence plot of the loan age variable on the predicted PD so that we can visualize how age is being used in the model.



Cluster-Robust Inference

As mentioned in the overview, we will use the cluster-robust covariance estimator for standard errors, which is why we haven't been using the regular standard errors from the model summary.

The heat map below shows the relative growth in the standard errors (SEs) going from the regular SEs (via MLE; inverse information) to the cluster-robust SEs (via sandwich estimator).



The SEs for most parameters grow by a fairly small amount (about 0% to 15%) and the SE for the estimated VTL ratio is inflated the most (about 45%). The SEs for credit score, the DTI missing indicator, the spline terms for age of loan, and the occupancy status parameters also increase a decent amount (about 20% to 30%).

It's not easily visible on the plot, but there are also a few SEs which decrease by a very small amount.

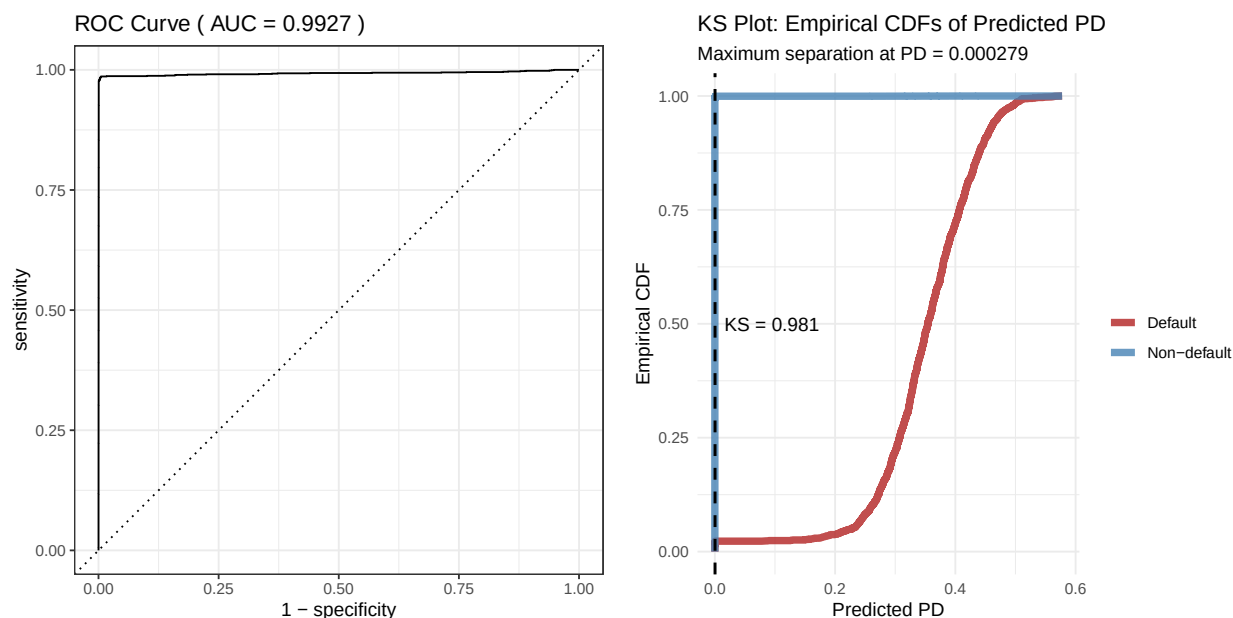
Overall, the standard errors do not increase substantially, suggesting that accounting for within-loan dependence does not materially alter the precision of the estimated effects. Nevertheless, the increase is large enough to warrant the use of cluster-robust standard errors for inference. With these more reliable uncertainty estimates, we report the confidence interval for a selected parameter below.

We estimate that a 10 point increase in credit score is associated with a 2.5559% relative growth in the odds of default next month with 99% confidence interval (2.5554%, 2.5565%) while adjusting for the selected macroeconomic variables and loan-specific variables.

Prediction Evaluation

We now test the model selected from earlier on the “out-of-time” (test set) data.

We do not report accuracy since our event is rare so this can be very high even if we predict all events wrong.



The ROC curve is highly bowed to the top-left corner and we have a very high AUC of about 99%, which indicates that the classifier performs very well on the test data.

The Kolmogorov-Smirnoff plot shows very large maximum separation (KS statistic) of about 98%, suggesting that the model separates the classes well.

Since our outcome is a rare event, we specifically look at the evaluation metrics when we use the empirical event rate from the training data as the threshold.

Confusion Matrix (Rows = Predicted, Columns = Actual)

	No Default (0)	Default (1)
No Default (0)	5725024	6080
Default (1)	35	1492

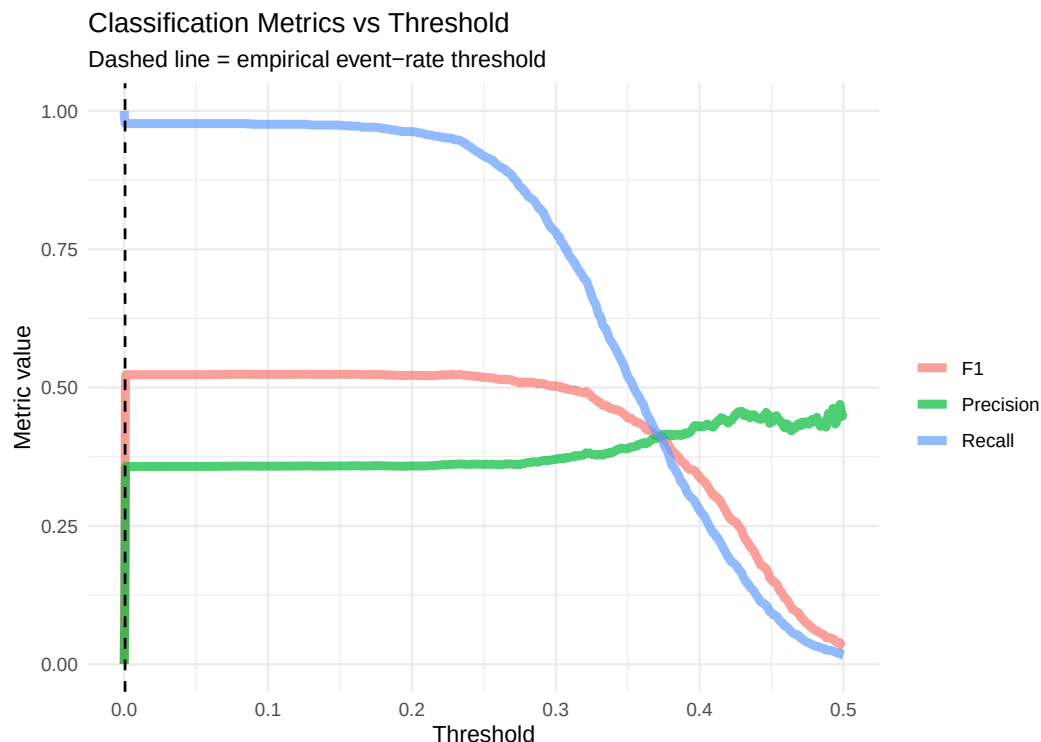
Other Classification Metrics

Sensitivity	Precision	F1
0.9770792	0.1970417	0.3279481

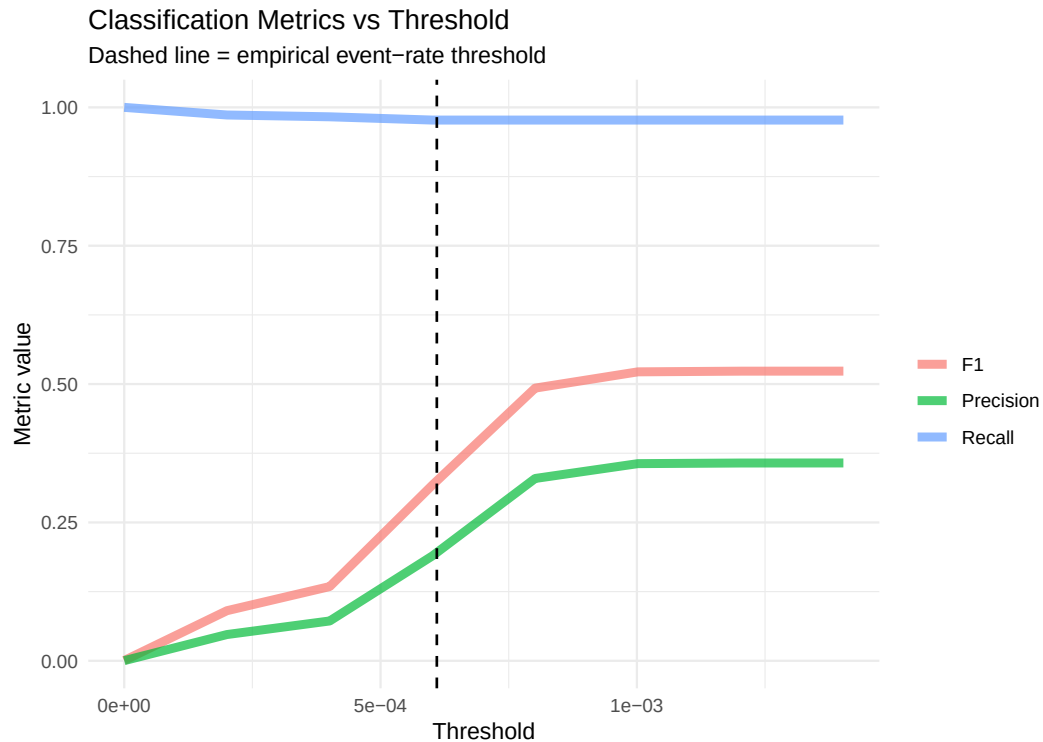
For the empirical event-rate threshold:

- The sensitivity (recall) value tells us that the model correctly classifies about 98% of actual defaults. Given that we are predicting rare events, this is one of the metrics we're most interested in, because failing to predict a default when it actually happens is likely much more costly than predicting one when it doesn't happen. That is, we care more about the true positive rate than the false positive rate.
- The precision value tells us that only about 20% of predicted defaults actually end up defaulting. As we just mentioned, precision is relatively less important for our application, although this is a very low value.
- The F1 score of about 0.3 reflects what we saw in the two metrics above; the model does well on recall but has quite poor precision.

Now we look at how these metrics behave under a continuum of thresholds:

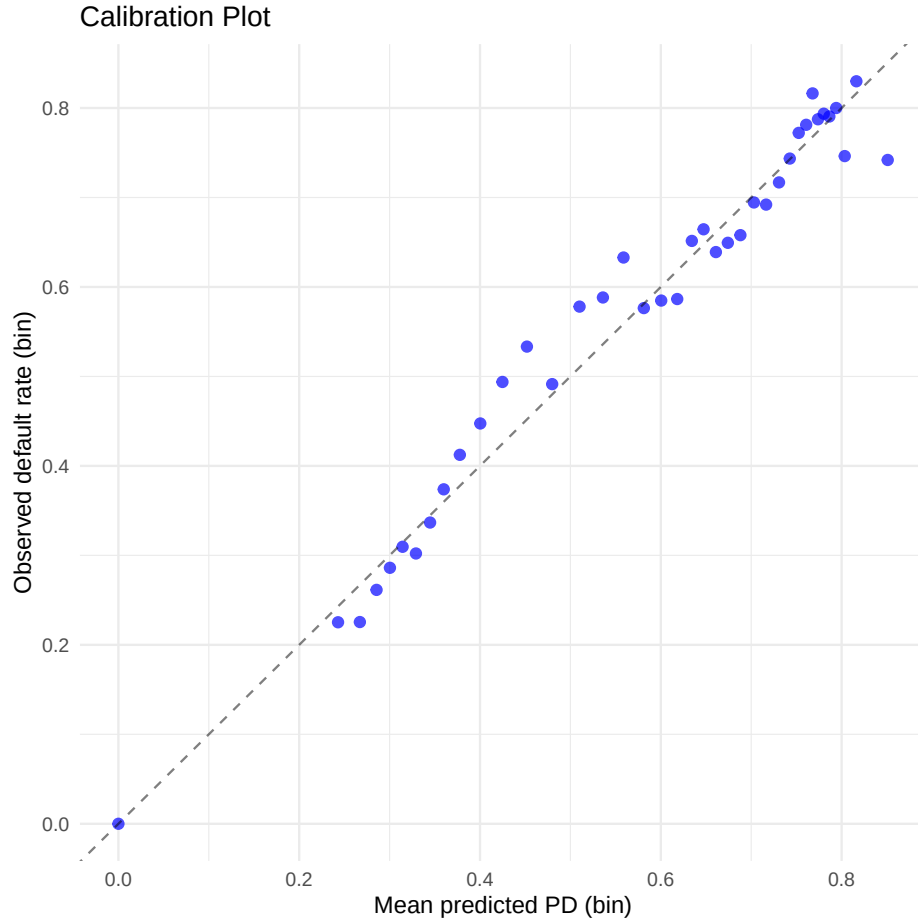


We zoom in on this plot for threshold values less than 0.0015 so that we can see that we can get a better visual of where exactly F1 is maximized.



We see that we can sacrifice a small amount of recall to gain a large improvement in recall by increasing our threshold to a value around 0.001 (which is a little less than double the empirical event-rate). At such a value for the threshold, we can achieve an F1 score as high as about 0.51.

We now assess calibration on the training data.



The binned predictions and observed default rates generally lie along the straight line through the origin with small variation, which suggests that the model fits the training data well.

Limitations and Possible Improvements

- **Aggregation and stress testing.** The current model focuses on loan-level next-month PD. An alternative or complementary approach would model aggregated default counts directly (e.g. via Poisson or quasi-Poisson regression), which may be more suitable for portfolio-level stress testing and scenario analysis.
- **Data limitations and sampling bias.** The analysis relies on the Freddie Mac Single-Family Loan-Level Dataset, which reflects agency-specific underwriting standards and may not be fully representative of the broader mortgage market.
- **Competing risks.** Prepayment is treated as censoring in the current framework. A more refined approach would explicitly model prepayment as a competing risk rather than assuming non-informative censoring.
- **Definition of default events.** Default is currently treated as an absorbing state. Future work could relax this assumption by allowing loans to re-enter non-default states and incorporating indicators or counts of prior defaults as covariates.
- **Feature set expansion and dynamic covariates.** Several potentially important extensions

were not explored, including:

- additional interaction terms motivated by economic intuition;
 - alternative constructions of borrower leverage, such as dynamic LTV based on modeled home price evolution;
 - approximate post-origination DTI measures using income and debt indices;
 - additional macroeconomic and policy uncertainty indicators.
- **Class imbalance.** Defaults are rare events in short-horizon prediction. While the current approach relies on likelihood-based estimation, alternative weighting or sampling strategies could be explored, with careful attention to calibration and interpretability.
 - **Variable selection and validation.** Variable selection was guided by economic reasoning and empirical diagnostics. A more formal procedure could incorporate L1-penalized regression (LASSO) and cross-validation on a dedicated validation set, while preserving a strictly held-out test period for final evaluation.
 - **Alternative modeling frameworks.** While logistic regression was chosen for interpretability and alignment with discrete-time hazard modeling, future work could benchmark against mixed-effects logistic models to capture unobserved loan heterogeneity, as well as more flexible neural networks or tree-based approaches as challengers rather than replacements.